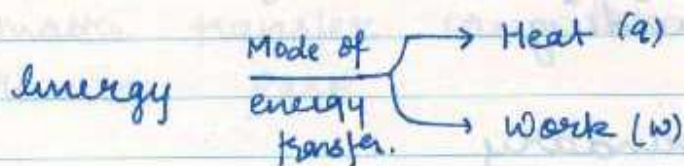


Thermodynamics

- It is branch of chemistry which deals with exchange of energy b/w system & surr. and the effects of that change on system parameters.
- It tells us about feasibility of a process.
- It tells us about extent of rxn.

Limitation :

- It does not involve time variable.
- Nuclear rxns are not part of Thermodynamics
- It does not deal with mechanism of rxn.



• Heat (q) $\Rightarrow$ It is process of energy transfer due to temperature difference or it is energy in transit.

• Work (w) $\Rightarrow$ It is process of energy transfer due to pressure difference.

* Some Basic Definitions $\Rightarrow$

1] System : It is part of universe which is under thermodynamic consideration. It can be as small as 1ml of water or it can be

as large as Indian ocean.

2) Surrounding : Everything around system, except system. It is always considered as infinitely large.

Note : Any process can be slow or fast, reversible or irreversible w.r.t system. But for surrounding, it is always considered as reversible.

System + Surrounding = Universe.

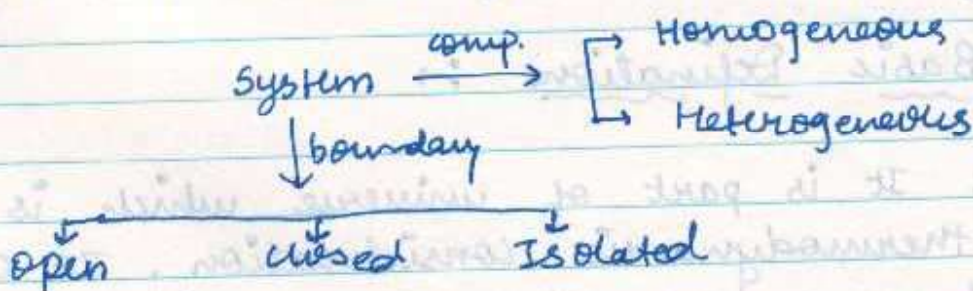
3) Boundary : It is the layer which separates system from surrounding.

Types :

- a) Real or imaginary boundary
- b) Rigid or flexible
- c) Permeable or impermeable
- d) Diathermic or Adiabatic

• Types of systems :

System can be classified on following basis :



On the basis of boundary,

- Open system : In it energy as well as matter transfer takes place. For eg, an open cup of tea.
- Closed system : In it energy transfer can take place but not matter transfer. For eg, a cup of tea covered with a plate.

Note : In thermodynamics all laws are applied only for closed system.

- Isolated system : In it neither energy nor matter transfer can take place. For eg, tea in thermos flask.

On the basis of composition,

Homogeneous

- If system is having uniform composition and phase throughout then it is called homogeneous. For eg, sugar solⁿ.

Heterogeneous

- A system having non uniform composition is called heterogeneous. For eg, chalk powder solⁿ.

- Q. A closed system cannot interact with sur. F
- Q. Atlantic Ocean is an open system. T
- Q. A homogeneous system must be pure. F
- Q. A pure system must be homogeneous. F

8. Fill in the blanks.

- 1) A vessel having perforated walls is open system.
 - 2) An adiabatic vessel fitted with adiabatic piston is closed system.
 - 3) An ad. rigid container is called isolated system.
- universe is considered as isolated system.

* Thermodynamic Properties :

Any directly or indirectly measurable quantity is called property of system. For eg, pressure, volume, temperature, concentration.

These are of two types :

- a) Intensive
- b) Extensive

Intensive

- These are the prop. which on division of system remains unchanged.
- eg: Density, temperature, etc.

- They are non-additive.

Extensive

- These are the prop. which get changed on division of system.
- eg: mass, volume, mole, etc.

- They are additive.

$$\boxed{\begin{array}{l} 1 \text{ M NaCl} \\ 100 \text{ g} \end{array}} + \boxed{\begin{array}{l} 2 \text{ M NaCl} \\ 200 \text{ g} \end{array}} \longrightarrow \boxed{\begin{array}{l} 3 \text{ M NaCl} \\ 300 \text{ g} \end{array}}$$

- generally ratio of two extensive qty. is intensive

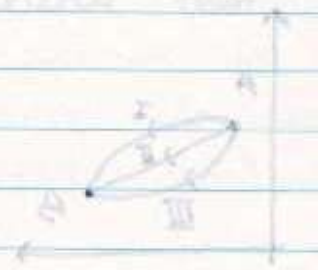
- generally product of intensive & extensive qty. is extensive.

Properties

I/E

P	I
V	E
T	I
n	E
H	E
U	E
G	E
S	E
A (Helmholtz Free Energy)	E
Vapour pressure	I
* EMF	I
Resistance	E
Boiling Point	I
Surface tension	I
Refractive index	I
Heat capacity	E
Molarity	I
Volume strength	I
Concentration	I

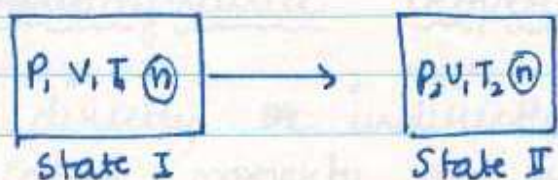
- If extensive qty. are divided by amount then they become intensive. For eg, molar heat capacity, specific volume, specific gravity



• Heat (q), Work (w) & time (t) are neither intensive nor extensive.

* State Parameters : These are properties or parameters which depend only on initial & final state of system.

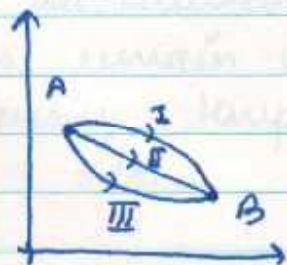
If out of given parameters even a single parameter changes, then we call it change of state.



Note : A state parameter like P, V, T, U, H, G, S should characterise state of a system.

* Path Parameters : These are parameters which depend on path followed when system changes its state.

For eg, work, heat, heat capacity, etc.



$$\Delta H_I = \Delta H_{II} = \Delta H_{III} = H_B - H_A$$

$$\Delta U_I = \Delta U_{II} = \Delta U_{III} = U_B - U_A$$

$$q_I \neq q_{II} \neq q_{III}$$

$$w_I \neq w_{II} \neq w_{III}$$

Note : Here U & H are state f.n. ΔU & ΔH are not state f.n. These simply represent change when state of system is changed.

* conditions for a function to be state f.n. $\Rightarrow$

i] if ϕ is a state f.n, then,

$$\int_A^B d\phi = \phi_B - \phi_A$$

ii] if ϕ is a state f.n, then,

$$\oint d\phi = 0$$

iii] A state f.n should characterise state of a system. For eg, H, U, G, S , etc., but $\Delta H, \Delta U, \Delta G, \Delta S$ are not state f.n. they are path independent parameters.

iv) if $\phi = f(x, y)$ is a state parameter then it should satisfy Euler's theorem

$$\left\{ \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) \right\}_x = \frac{\partial}{\partial x} \left\{ \left(\frac{\partial \phi}{\partial y} \right) \right\}_y$$

eg P.T. $f(x, y) = x^2 y + x y^2$ is a state f.n.

$$\rightarrow 2x + 2y = 2x + 2y$$

v] If $\Phi = f(x, y)$ is a state fun. then,
it can be written as

$$d\Phi = \left(\frac{\partial \Phi}{\partial x}\right)_y dx + \left(\frac{\partial \Phi}{\partial y}\right)_x dy$$

eg,

* $H = f(P, T)$

$$dH = \left(\frac{\partial H}{\partial P}\right)_T dP + \left(\frac{\partial H}{\partial T}\right)_P dT$$

* $U = f(V, T)$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

Q. Prove that pressure is intensive property.

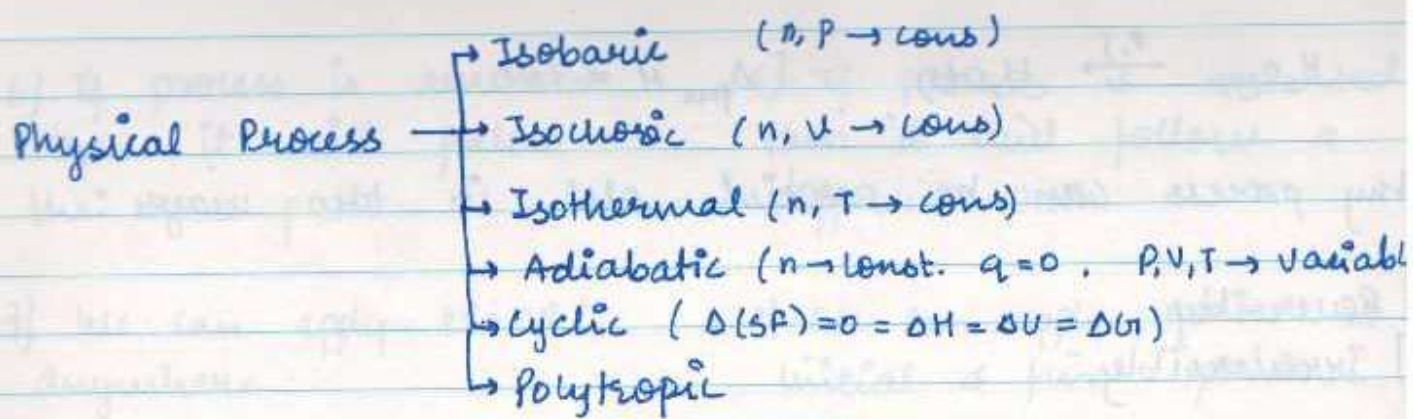
$$\rightarrow P = \frac{nRT}{V} \rightarrow \frac{F}{V} \rightarrow \frac{F}{L^3}$$

• Process $\rightarrow$ It is the path followed when system changes its state.

Types of process :

- Physical process
- Chemical process

Physical process : These are process during which chemical identity of system remains unchanged.
For eg. melting, vapourisation, etc.



• During isobaric process pressure remains constant throughout

Q. If initial & final temp. is same then process must be isothermal.

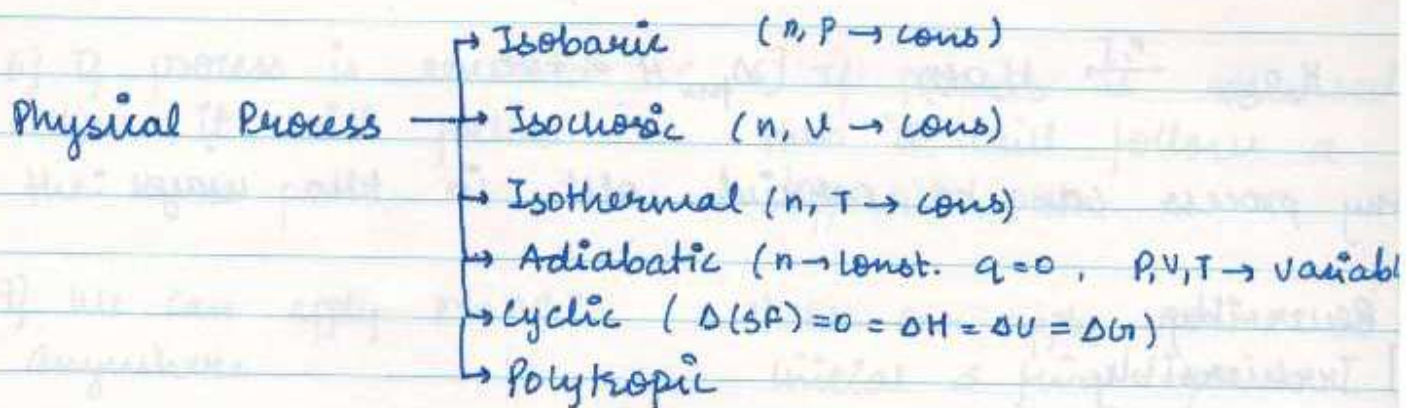
→ False.

* Chemical Process :

During it one type of substance gets converted into chemically other type of substance. During a chemical rxn temp. is always kept constant and out of pressure & volume one param will be constant at a time.



Note : Although phase transformations are physical processes but still we will apply all laws for phase transformation which we use for chemical rxn.



• During isobaric process pressure remains constant throughout

Q. If initial & final temp. is same then process, must be isothermal.

→ False.

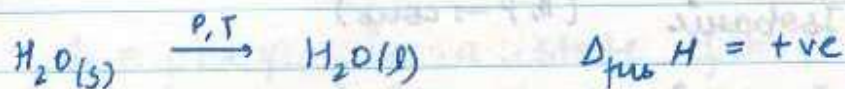
* Chemical Process :

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Note : Although phase transformations are physical processes but still we will apply all laws for phase transformation which we use for chemical rxn.



Any process can be carried out in two ways:

- i] Reversibly
- ii] Irreversibly

Reversible Process



- 2) In it external forces change gradually.

- 3) This process will be slow (Hypothetical)

- 4) Int. pressure changes at every instant.

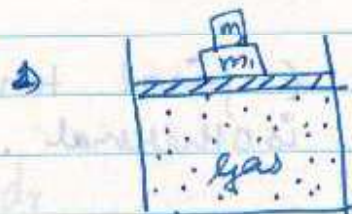
$$P_{\text{gas}} = P_{\text{ext}} \pm dp^{\circ}$$

$$P_{\text{gas}} \approx P_{\text{ext}}$$

- 5) Here we can plot



Irreversible Process



- 2) External forces change suddenly

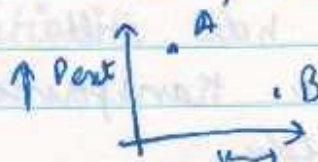
- 3) This process will be fast (real)

- 4) Here process takes place at constant pressure

$$P_{\text{gas}} = P_{\text{ext}} \pm \Delta P$$

$$P_{\text{gas}} \neq P_{\text{ext}}$$

- 5) We cannot plot P_{gas} v/s V so we plot P_{ext} v/s V



6] If process is reversed then it will follow the same path.

6] If process is reversed then it will follow a different path.

7] We can apply $PV = nRT$ anywhere

7] We can apply $PV = nRT$ at initial & final points only.

8] In reversible cyclic process system as well as surrounding are restored to their original pos.

8] In irreversible cyclic process system is restored to its original pos. but not surrounding

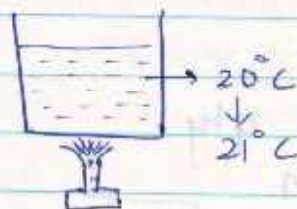
Note : A process can be reversible or irreversible w.r.t system but it is always considered as reversible for surrounding.

* Heat capacity : Amt of heat req. to raise the temp of 1 gm of substance by 1°C .

$$C = \frac{q}{\Delta T} \quad (\text{J/K})$$

where,

$C \rightarrow$ heat capacity
 $q \rightarrow$ heat exchanged
 $\Delta T \rightarrow$ Change in temp.



- It is extensive qty.

- C is a path function.

* Molar Heat Capacity (C_m) : Amount of heat req. to raise the temp. of one mole of substance by 1°C .

$$C_m = \frac{q}{n \Delta T} \quad (\text{J/K-mol})$$

where,

$C_m \rightarrow$ molar heat capacity

$q \rightarrow$ heat exchanged

$\Delta T \rightarrow$ change in temp.

$n \rightarrow$ no. of moles

- It is an intensive qty

- Path f'n

* Specific Heat Capacity (C_s) : Amount of heat req. to raise the temp. of one gm of substance by 1°C .

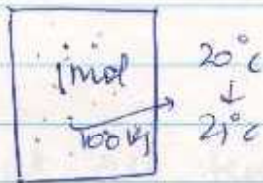
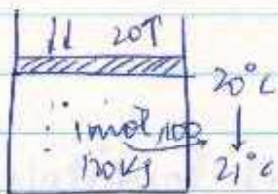
$$\frac{C}{m} = C_s = \frac{q}{m \Delta T} \quad (\text{J/K-kg})$$

- Intensive qty

- Path f'n

* In case of gases heat capacities are of two types,

- i) molar heat capacity at const pressure ($C_{p,m}$)
- ii) " " " " volume ($C_{v,m}$)



$$C_{p,m} > C_{v,m}$$

- State f.n
- Intensive qty.

- For an ideal gas,

$$C_p - C_v = nR$$

$$\frac{C_p}{n} - \frac{C_v}{n} = R$$

$$C_{p,m} - C_{v,m} = R$$

$$\frac{C_{p,m}}{C_{v,m}} = \gamma = \text{adiabatic coeff}$$

$$\frac{C_{p,m}}{C_{v,m}} - 1 = \frac{R}{C_{v,m}}$$

$$\gamma - 1 = \frac{R}{C_{v,m}}$$

$$C_{vm} = \frac{R}{\gamma - 1}$$

$$C_{pm} = \frac{\gamma R}{\gamma - 1}$$

For ideal gas,

$$C = f(T) \text{ only}$$

$$C = a + bT + cT^2 + \dots$$

For solids & liquids,

$$C_{p,m} \approx C_{v,m} \approx C_m$$

* Heat capacities for various processes :

1] For Isobaric process

$$C_p = \frac{q_p}{\Delta T}$$

2] For Isochoric process

$$C_v = \frac{q_v}{\Delta T}$$

3] For adiabatic process

$$C = \frac{q}{\Delta T} = 0$$

4] For isothermal process

$$C = \frac{q}{\Delta T} = \infty$$

* Degree of Freedom $\rightarrow$

It represents the no. of ways thru which a system can change from its current state.

Types:

- 1) Translational (KE) $\rightarrow$ at all temp.
- 2) Rotational (KE) $\rightarrow$ at moderate temp.
- 3) Vibrational (KE+PE) $\rightarrow$ very high temp.

$$\text{Total DOF} = 3N$$

$N \rightarrow$ no. of atoms in a given molecule.

Q. Find total DOF for

a) He $\rightarrow 3$

b) $\text{H}_2 \rightarrow 6$

c) $\text{CO}_2 \rightarrow 6$

d) $\text{H}_2\text{O} \rightarrow 9$

e) $\text{NH}_3 \rightarrow 12$

f) $\text{C}_2\text{H}_6 \rightarrow 24$

Translational 3 (for all)

Rotational 0 (Mono)

2 (Di, Tri (linear))

3 (Poly)

Vibrational 0 (Mono)

$3N-5$ (Di, Tri linear)

$3N-6$ (Poly)

* law of equipartition of energy:

Accⁿ to it, energy is distributed among various degrees of freedom equally.

for translational DOF

Energy associated with each trans DOF = $\frac{1}{2} kT$ J/molecule
per molecule

Energy associated " " " " = $\frac{1}{2} RT$ J/mol
per mole

If translational DOF are f_T

Then total energy associated with f_T no. of
trans DOF per mole = $f_T \frac{RT}{2}$ J/mol.

for Rotational DOF

Same as above.

for Vibrational DOF

Energy associated with each vibration DOF = kT J/molecule
per molecule

" " " " " " = RT J/mol
per mole

• Internal energy for 1 mol,

$$U = f_t \frac{RT}{2} + f_R \frac{RT}{2} + f_{vb} RT$$

$$U = (f_t + f_R + 2f_{vb}) \frac{RT}{2}$$

$$C_{V,m} = \left(\frac{f_t}{2} + \frac{f_R}{2} + f_{vb} \right) R$$

$$C_{P,m} = C_{V,m} + R$$

$$\gamma = \frac{C_{P,m}}{C_{V,m}}$$

g. Find out $C_{P,m}$, $C_{V,m}$ & γ for

	$C_{P,m}$	$C_{V,m}$	γ
a) He (g) $\Rightarrow$	$5R/2$	$3R/2$	$5/3$

b) H_2 (g) $\Rightarrow$	$7R/2$	$5R/2$	$7/5$
----------------------------	--------	--------	-------

c) CO_2 (g) $\Rightarrow$	$7R/2$	$5R/2$	$7/5$
-----------------------------	--------	--------	-------

d) H_2O (g) $\Rightarrow$	$4R$	$6R/2$	$4/3$
-----------------------------	------	--------	-------

e) NH_3 (g) $\Rightarrow$	$3R$	$3R$	$4/3$
-----------------------------	------	------	-------

* Atomicity	f_t	f_R	f_{vib}	C_V		C_P		γ	
				ex. vib	inc. vib	ex. vib	inc. vib	ex. vib	inc. vib
Mono	3	0	0	$\frac{3R}{2}$	$3R/2$	$5R/2$	$5R/2$	$5/3$	$5/3$
Di	3	2	1	$5R/2$	$3R$	$7R/2$	$4R$	$7/5$	$4/3$
Tri	linear	3	2	$5R/2$	$9R/2$	$7R/2$	$11R/2$	$7/5$	$11/9$
	Non Lin	3	3	$3R$		$4R$		$4/3$	

Note: As atomicity $\uparrow$, gamma decreases

* Parameters Related to 1st law of Thermodynamics

1] Work (W): It is transfer of energy due to pressure difference.

- Neither intensive nor extensive
- It is path f:n

Types of Work :-

- PV or expansion work
- Non PV work (eg. muscle work, electrical work)

In first law, we will mainly deal with PV or expansion work.

Sign convention :-

- Work done by system or work of expansion is taken as -ve.
- Work done on the system or work of compression is taken as +ve.

$$W = -20 \text{ J}$$

$\Rightarrow$ work done by the system = 20 J.

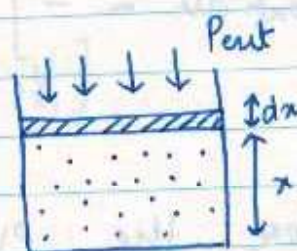
Calculation of Work $\Rightarrow$

$$|dw| = |F_{\text{ext}} dx|$$

$$|dw| = \left| \frac{F_{\text{ext}}}{A} \cdot A dx \right|$$

$$|dw| = |P_{\text{ext}} \cdot dv|$$

$$|dw| = - \int P_{\text{ext}} dv$$



$$\Rightarrow W = - \int_{V_1}^{V_2} P_{\text{ext}} dv \rightarrow \text{Applicable for all kind of processes \& for all types of fluid in closed vessel.}$$

1) If $P_{\text{ext}} = 0 \Rightarrow$ Free expansion will take place

$$W = - \int P_{\text{ext}} dv = 0$$

2) If process is irreversible

$$P_{\text{ext}} = \text{constant}$$

$$W = -P_{\text{ext}} \int_{V_1}^{V_2} dV$$

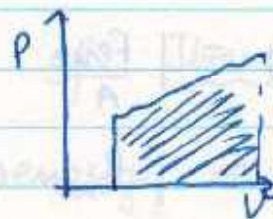
$$\Rightarrow W = -P_{\text{ext}} (V_2 - V_1)$$

3) If process is reversible

$$P_{\text{ext}} \approx P_{\text{gas}} = \frac{nRT}{V}$$

$$W = - \int P_{\text{ext}} dV = - \int_{V_1}^{V_2} nRT \frac{dV}{V}$$

4) area under the PV curve will give magnitude of work done.



5) If $P = f(V)$ eg $P = a + bV$

$$W = - \int_{V_1}^{V_2} (a + bV) dV$$

Q. Calculate work done for exp. of a substance
from 3m^3 to 5m^3 against

a) Constant Pressure = $10^5 \text{ Pa} \Rightarrow -10^5 (2) \text{ J}$

b) Variable Pressure = $(10 + 5V) \text{ Pa} \Rightarrow -(10 \times 2 + 40) \text{ J}$

c) Free exp. $\Rightarrow 0$

d) For 1 mol of ideal gas following $T = 4V^2$
 $\Rightarrow -nR \left(\frac{V_2^2 - V_1^2}{2} \right)$

9. Derive an expression for work done for n mol of Vanderwall gas. (constant temp.)

$$\rightarrow (P + \frac{an^2}{V^2})(V - nb) = nRT \quad \Rightarrow \quad P = \frac{nRT}{(V - nb)} - \frac{an^2}{V^2}$$

$$W = - \int P dV$$

$$\Rightarrow W = - \left[\int_{V_1}^{V_2} \frac{nRT}{(V - nb)} dV + \int_{V_1}^{V_2} \frac{an^2}{V^2} dV \right]$$

$$W = -nRT \ln \frac{V_2 - nb}{V_1 - nb} - an^2 \left[\frac{1}{V_2} - \frac{1}{V_1} \right]$$

Unit conversion :

$$1 \text{ bar L} = 100 \text{ J}$$

$$1 \text{ atm L} = 101.3 \text{ J}$$

* Heat (q) :

It is transfer of energy due to temp. difference

- It is neither extensive nor intensive
- It is path f'n.

Sign convention :

- Heat lost by the system is taken as -ve
- Heat gained by system is taken as +ve

Heat exchanged at constant volume (q_v) = ΔU
 " " " " Pressure (q_p) = ΔH

$$dq = n C_m dT$$

$$dq_v = n C_{v,m} dT$$

$$\Rightarrow \frac{dq_v}{dT} = n C_{v,m} = C_v$$

$$\left(\frac{\partial U}{\partial T} \right)_v = n C_{v,m} = C_v$$

$$||^4 \left(\frac{\partial H}{\partial T} \right)_p = n C_{p,m} = C_p$$

3] Internal Energy (U or E) $\Rightarrow$

It is sum of all kind of energies present in the system.

$$U = KE + PE + \underbrace{BE + \dots}_{\text{negligible effect of change in } P, V, T}$$

We cannot determine absolute value of U

- It is extensive qty.
- State f.n.

In thermodynamics, we consider system at rest & except gravitational fields, all other fields are assumed to be absent.

If Internal Energy is changed,

$$\Delta U = \Delta KE + \Delta PE$$

In case of ideal gas there are no interactions b/w particles $\Rightarrow PE = 0$

$$\Delta U = \Delta KE$$

For an ideal gas, $KE \propto f(T)$ only

$$U = f(T)$$

For an ideal gas, U is a f'n of temp. only
For real gas & other substances

$$U = f(P, V, T)$$

In case of gases, if eqⁿ of state is known then state parameters can be determined using two variables at a time.

$$U = f(P, V) = f(P, T) = f(V, T)$$

For an ideal gas,

$$\left(\frac{\partial U}{\partial V}\right)_T = 0$$

$$\left(\frac{\partial U}{\partial P}\right)_T = 0$$

$$\left(\frac{\partial U}{\partial T}\right)_V = n C_{V,m} = C_V$$

We know that U is a state f:n

$$\left. \begin{aligned} dU &= \left(\frac{\partial U}{\partial P}\right)_V dP + \left(\frac{\partial U}{\partial V}\right)_P dV \\ dU &= \left(\frac{\partial U}{\partial P}\right)_T dP + \left(\frac{\partial U}{\partial T}\right)_P dT \\ dU &= \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT \end{aligned} \right\} \text{Applicable only for Physical process}$$

All three differentiable eqⁿ are correct but we will use third eqⁿ extensively.

Calculation of ΔU for Physical Process:

1) For an ideal gas undergoing any process

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

$$\Rightarrow \boxed{\int dU = n \int C_{V,m} dT}$$

if $C_{V,m}$ is independent of temp.,

$$\boxed{\Delta U = n C_{V,m} (T_2 - T_1)}$$

2) Real gas undergoing isochoric process

$$dU = 0$$

$$\int_{U_1}^{U_2} dU = n \int_{T_1}^{T_2} C_{V,m} dT$$

3) For solids & liquids.

Change in volume is negligible.
 $dv \cong 0$

$$\int_{U_1}^{U_2} du = n \int_{T_1}^{T_2} c_{m,dT}$$

4) For a real gas undergoing any process

$$du = n c_{v,m} dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

Q. Calculate ΔU for a gas undergoing change of state from state I ($300K, 2 \times 10^{-2} m^3$) to state II ($400K, 4 \times 10^{-2} m^3$) for 1 mol of gas.

a) If gas is ideal

b) If gas is van der Waals

$$\left\{ \text{Equien : } \left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_V - P \right.$$

$$\left. \begin{aligned} c_{v,m} &= 12 J K^{-1} mol^{-1} \\ a &= 2 J.m. / mol^2 \end{aligned} \right\}$$

$$\rightarrow a) \Delta U = n c_{v,m} (100) = 1200 J$$

$$P + \frac{an^2}{V^2} = \frac{nRT}{(V-nb)}$$

$$\begin{aligned} b) \int du &= n c_{v,m} (T_2 - T_1) + \int_{0.02}^{0.04} \left(T \left(\frac{\partial P}{\partial T} \right)_V - P \right) dV \\ &= 1200 + \int_{0.02}^{0.04} \left(\frac{nRT}{V-nb} - P \right) dV = \end{aligned}$$

* First law of Thermodynamics :

This law is based on conservation of energy. Accⁿ to it,

Energy can neither be created nor destroyed, it can just be transferred from one body to another.

- Suppose a system has internal E 'U,' it exchanges some heat with surrounding & W.D. is 'W', so,

$$U_2 = U_1 + q + W$$

$$\Delta U = q + W$$

or

$$du = dq + dw$$

} Applicable for all kinds of process & for all types of fluids in closed system

Q. The pressure of a fluid is a f.n of volume $P = a + bV$ & IE is given by $U = 34 + 3PV$. Find a, b, W, ΔU & q for change in state from $(100 \text{ Pa}, 3 \text{ m}^3) \rightarrow (400 \text{ Pa}, 6 \text{ m}^3)$

$$\rightarrow \Delta U = 7234 - 934 = 6300$$

$$a + 3b = 100 \quad \& \quad 6b + a = 400 \Rightarrow b = 100, a = -200$$

$$W = - \int (-200 + 100V) dV$$

$$\Rightarrow W = -700$$

$$q = 7050.$$

* ENTHALPY :-

- represented by 'H'
- Mathematically it is defined as,

$$H = U + PV$$

- Intensive property of state f.n
- Absolute value of 'H' cannot be determined.

$$H = U + PV$$

$$H + dH = U + du + (P + dp)(V + dV)$$

$$dH = dq + dw_{pv} + dw_{non-pv} + PdV + VdP$$

$$dH = dq + dw_{non-pv} + VdP$$

At constant pressure,

$$dH = dq_p + dw_{non-pv}$$

if $dw_{non-pv} = 0$

$$dH = dq_p \quad \text{or} \quad \Delta H = q_p$$

Note : $\Delta H = \Delta U + (P_2V_2 - P_1V_1)$

Calculation of ΔH ,

a) For physical process,

$$H = U + PV$$

$$\Delta H = \Delta U + \Delta PV$$

For an ideal gas

$$H = U + nRT$$

$$H = f(T)$$

For real gas & other subs.

$$H = f(P, V, T)$$

$$H = f(P, V) = f(V, T) = f(P, T)$$

For ideal gas,

$$\left(\frac{\partial H}{\partial P}\right)_T = 0 \quad \left(\frac{\partial H}{\partial V}\right)_T = 0 \quad \left(\frac{\partial H}{\partial T}\right)_P = nC_{p,m} = C_p$$

$\therefore H$ is a state f:n,

$$dH = \left(\frac{\partial H}{\partial P}\right)_V dP + \left(\frac{\partial H}{\partial V}\right)_P dV \quad \text{--- (I)}$$

$$= \left(\frac{\partial H}{\partial V}\right)_T dV + \left(\frac{\partial H}{\partial T}\right)_V dT \quad \text{--- (II)}$$

$$= \left(\frac{\partial H}{\partial P}\right)_T dP + \left(\frac{\partial H}{\partial T}\right)_P dT \quad \text{--- (III)}$$

Out of given eqⁿ. we will use (III) eq. extensively.

• Case I : For an ideal gas undergoing any process.

$$dH = \left(\frac{\partial H}{\partial P}\right)_T dP + \left(\frac{\partial H}{\partial T}\right)_P dT$$

$$\Delta H = n \int_{T_1}^{T_2} C_{p,m} dT$$

- Case II : For real gas undergoing isobaric process.

$$dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP^0$$

$$\Delta H = n \int_{T_1}^{T_2} C_{p,m} dT$$

- Case III : A real gas undergoing any process

$$dH = n C_{p,m} dT + \left(\frac{\partial H}{\partial P}\right)_T dP$$

$$\Delta H = n \int_{T_1}^{T_2} C_{p,m} dT + \int_{P_1}^{P_2} \left(\frac{\partial H}{\partial P}\right)_T dP$$

Now using defⁿ,

$$\Delta H = \Delta U + \Delta(PV)$$

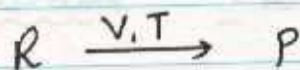
For an ideal gas,

$$\Delta H = \Delta U + nR\Delta T$$

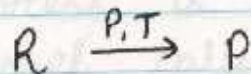
For solids & liquids,

$$\Delta H \cong \Delta U.$$

- Calculation of ΔH for chemical process.

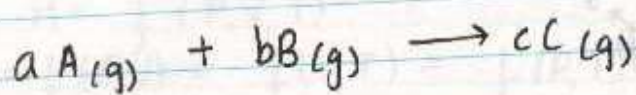


$$q_v = \Delta U$$



$$q_p = \Delta H$$

- Case I : if only gaseous reactants & products are present.



• We have assumed that all gases are ideal

$$\Delta H = \Delta U + (\Delta n_g) RT$$

- Δn_g = moles of gaseous product - moles of gas. reactants

- Case II : If along with gases, solids & liquids are also present



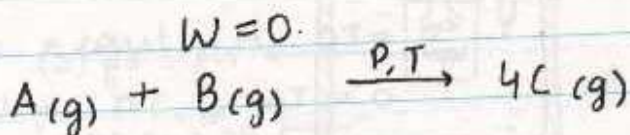
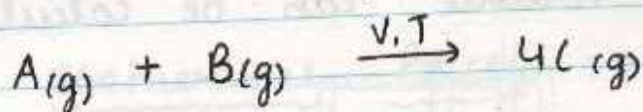
due to solids & liquids $\Delta(PV) \approx 0$

Q.

$$\Delta H = \Delta U + (\Delta n_g) RT$$

- $\Delta n_g = d - a$.

Work done during a chemical rxn:



$$W = - \int_{V_1}^{V_2} P dV = - P_{ext} [V_2 - V_1]$$

$$= - P_{ext} \left[\frac{n_1 RT}{P} - \frac{n_2 RT}{P} \right]$$

$$= \boxed{- \Delta n_g RT}$$

* CALORIMETRY:

It is branch of chemistry which deals with measurement of energy exchange during phys or chemical process coz it is carried out in vessel called calorimeter thats why it is called calorimetry.

Measurements are done under,

- i) At constant $V \& T$ (q_v or ΔU)
- ii) At constant $P \& T$ (q_p or ΔH)

A) At constant $V \& T$ $\therefore$

This process is carried out in a vessel closed rigid vessel called bomb calorimeter.

In this process small amount of substance undergoes combustion. As a result temperature

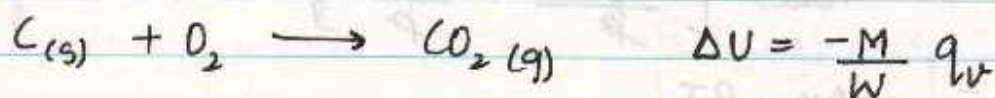
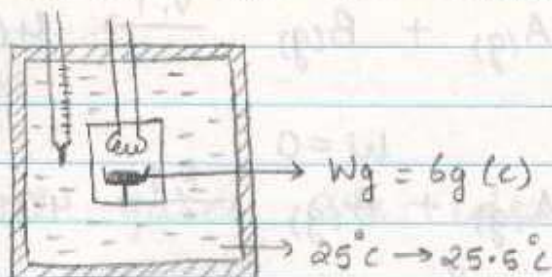
outside system (water + calorimeter) changes & heat exchanged at constant volume can be calculated using

$$q_v = C \Delta T$$

where,

q_v = heat exchanged at constant volume

C = Heat capacity of water + calorimeter



Bomb Calorimeter is generally used when gaseous components are involved.

B] At constant P & T $\Rightarrow$

generally rxns involving solids & liquids are carried out at constant P & T in open styrofoam cups.



$$q_p = C \Delta T (\Delta H)$$

• Calculation of q , w , ΔH & ΔU :-

1] For Isothermal Process :-

- $T, n = \text{const.} \Rightarrow \Delta T = 0$

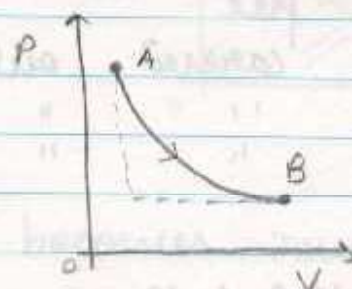
- $\Delta U = nC_{v,m} \Delta T = 0$

- $\Delta H = nC_{p,m} \Delta T = 0$

From FLT,

$$\Delta U = q + w$$

$$\Rightarrow \boxed{q = -w}$$



Magnitude of work will depend on whether process is rev. or irrev.

Case I For Reversible process,

$$W = - \int_{V_1}^{V_2} P dV = \boxed{-nRT \ln V_2/V_1}$$

$$= \boxed{-nRT \ln P_1/P_2}$$

Case II For irreversible process

$$\boxed{W = - \int_{V_1}^{V_2} P_{\text{ext}} dV = -P_{\text{ext}} (V_2 - V_1)}$$

Note : If P_{ext} is not given then final pressure of system is considered as ext. pressure

Q. One mole of an ideal gas is expanded isothermally at 300 K from 10 atm to 1 atm . Calculate q , w , ΔH & ΔU under following conditions

- i) If exp. is free
- ii) " " " carried out revers.
- iii) " " " " " "
- iv) " " " " " " against $P_{\text{ext}} = 0.5\text{ atm}$

→ ii) $w = -300R \ln 10$
 $q = 300R \ln 10$
 $\Delta H = 0$
 $\Delta U = 0$

$10 \times V_1 = R \times 300$
 $V_2 = 300R$

iii) $-270R = w$
 $270R = q$

iv) $w = -135R$
 $q = 135R$

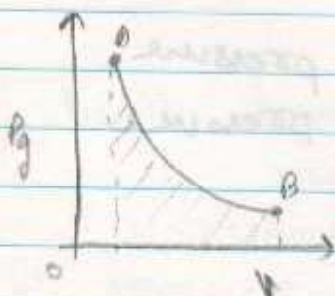
i) $w = q = \Delta H = \Delta U = 0$

free exp. on an ideal gas will be isothermal as well as adiabatic.

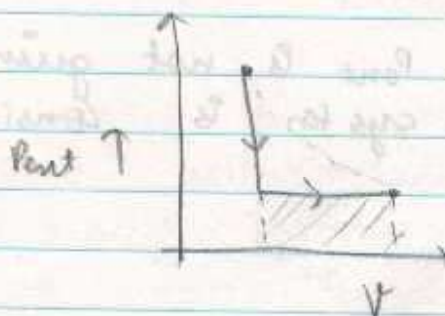
* Graphical analysis of work:
 (Isothermal Process)

For expansion

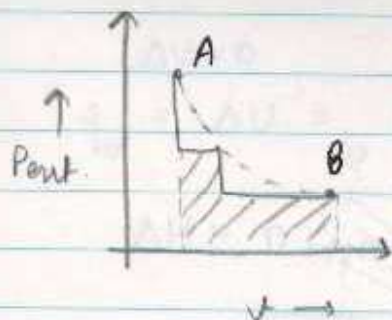
Reversible



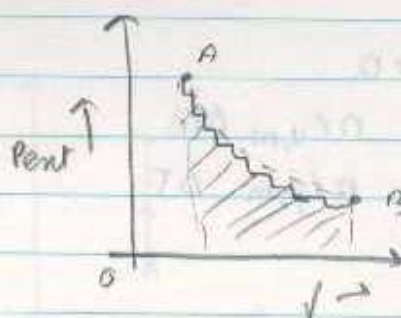
Irreversible (single stage)



Irreversible (Two stage)



Irreversible (n stages)



If $n \rightarrow \infty$ then process becomes reversible

$$|W|_{rev} > |W|_{irr, n \text{ stages}} > |W|_{irr, 1 \text{ stage}}$$

• Reversible processes are more efficient w.r.t work coz we have to do minimum work during compression & we can extract max. work during expansion from the system thru reversible process.

- Q. One mole of an ideal gas is expanded from (10 atm, 10L) to (2 atm, 50L) respectively. Then cal. 'w'.
- First against 5 atm & then against 2 atm
 - If expansion occur in single stage

→ a) $W = -5(10L) = -200 \times 101.3 \text{ J}$

b) $W = -80 \times 101.3 \text{ J}$

b)

c) $W = nRT \ln 2 + nRT \ln \frac{3}{2} + nRT \ln \frac{5}{3}$
 $= RT \ln 5 = 8.3 \times \frac{100}{0.0821} \times \ln 5$

Q. 2] For Isobaric Process $\rightarrow$

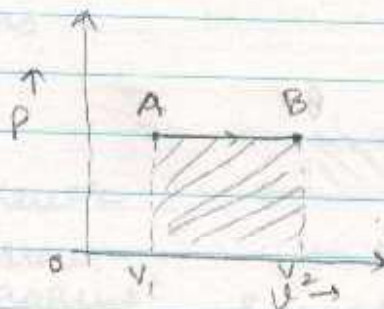
$$\Delta P = 0$$

$$\Delta U = n C_{v,m} \Delta T$$

$$\Delta H = n C_{p,m} \Delta T$$

From FLT,

$$\Delta U = q + w$$



$$w = - P_{ext.} (V_2 - V_1) = - n R \Delta T$$

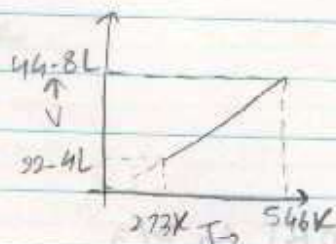
$$n C_{v,m} \Delta T = q_p - n R \Delta T$$

$$\therefore q_p = n C_{p,m} \Delta T$$

$$\frac{\Delta H}{\Delta U} = \frac{C_{p,m}}{C_{v,m}} = \gamma$$

Note: Magnitude of w.d. will be same irrespective of reversible or irreversible isobaric process.

Q. For one mole of an ideal gas calculate 'w'.



$$\rightarrow w = - R \cdot 273$$

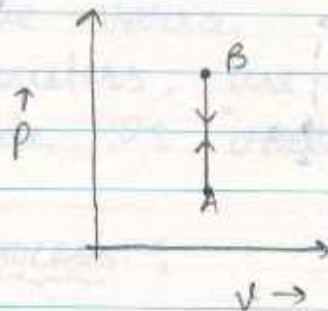
3] Isochoric Process :

$$\Delta V = 0$$

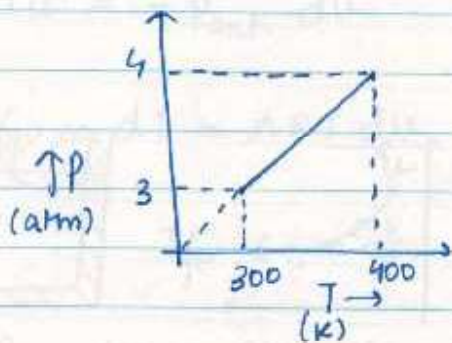
$$q_w = \Delta U = n C_{v,m} \Delta T$$

$$\Delta H = n C_{p,m} \Delta T$$

$$W = - \int P dV = 0.$$



Q. For one mole of a monoatomic ideal gas calculate $q, w, \Delta H, \Delta U$.



$$\rightarrow W = 0$$

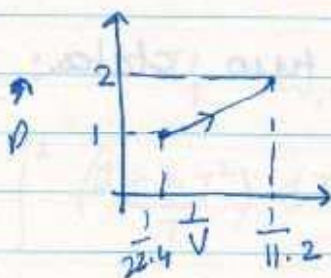
$$q_w = \Delta U = 1 \times \frac{3}{2} \times R \times 100$$

$$\Delta H = (1 + \frac{3}{2}) R \times 100$$

Q. Calculate w for an ideal gas
[$\ln 2 = 0.7$]

$$PV = nRT$$

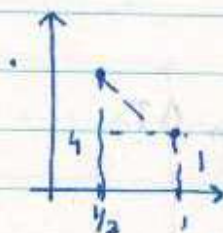
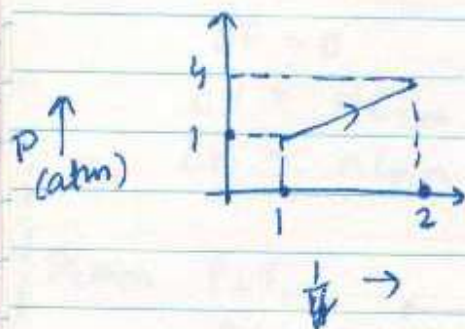
$$P = nRT \cdot \frac{1}{V}$$



$$w = -nRT \ln \frac{1}{2}$$

$$= \ln 2 [-22.4]$$

g. Calculate w [$\ln 2 = 0.7$]



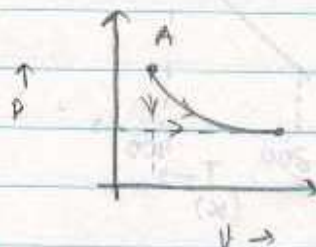
$$\rightarrow P = \frac{nRT}{V} = + \left(\frac{1}{2} \times \frac{1}{2} \times 3 + 3 \times \frac{1}{2} \right) \quad | \quad - \int P dV$$

4] For Adiabatic Process $\Rightarrow$

$$q = 0$$

$$\Delta U = n C_{v,m} \Delta T$$

$$\Delta H = n C_{p,m} \Delta T$$



$$w = \Delta U = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

$$W = \frac{nR\Delta T}{\gamma - 1} = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = \Delta U$$

Between two states of an ideal gas only one adiabatic process is possible.
Adiabatic processes are unique b/w two states.

$$P_1 V_1 T_1 \xrightarrow[\text{adiabatic}]{\text{rev.}} P_2 V_2 T_2$$

$$P_1 V_1 T_1 \xrightarrow[\text{adiabatic}]{\text{irrev.}} P_2 V_2 T_2$$

If we move from $A(P, V, T)$ to $B(P_2, V_2, T_2)$ reversibly then we can't achieve (P_2, V_2, T_2) by adiabatic irreversible process and vice versa. To calculate final parameters, we should know whether process is reversible or irreversible.

For reversible adiabatic process:

From FLT,

$$dU = dq + dw$$

$$dU = -P_{ext} dV$$

$$n C_{v,m} dT = nRT \frac{dV}{V}$$

Case I: If $C_{v,m}$ is independent of temp.

$$\frac{R}{\gamma-1} dT = RT \frac{dV}{V}$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_2}{V_1} \right)^{\gamma-1}$$

$$\Rightarrow TV^{\gamma-1} = k$$

$$\Rightarrow PV^{\gamma} = k$$

Case II: If $C_{v,m}$ is $f(T) = a + bT^2$

$$\int_{T_1}^{T_2} \left(\frac{a+bT^2}{T} \right) dT = -R \int_{V_1}^{V_2} \frac{dV}{V}$$

Q. For Irreversible process :-

$$du = dq + dw$$

$$n C_{v,m} (T_2 - T_1) = -P_{ext} (V_2 - V_1)$$

$$\Rightarrow \frac{nR(T_2 - T_1)}{\gamma - 1} = -P_{ext} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$$

Q. For 1 mol of non linear triatomic ideal gas is expanded adiabatically at 300K from 16 atm to 1 atm. Calculate work done

i) if expansion is free $\Rightarrow 0$

ii) " " is carried out rev. $\Rightarrow$

iii) " " " " " irrev.

iv) " " " " " against $P_{ext} = 0.5 \text{ atm}$

→ i) 0

$$ii) 16 \times V = 1 \times 0.0821 \times 300 \quad ; \quad \gamma = 1 + 2/6 = 4/3$$

$$\Rightarrow W = PV^\gamma = 16 \left(\frac{24.63}{16} \right)^{4/3}$$

$$-450R$$

$$T P^{\gamma-1} = K$$

$$T^{4/3} P^{1/3} = K$$

$$\frac{(T_2 P_2)^{1/3}}{P_2} = K \quad T_2^4 P_2 = T_1^4 P_1$$

$$iii) \frac{nR(-150)}{1/3} = -P_{ext} \left(\frac{nR(150)}{0.5} - \frac{nR(300)}{16} \right)$$

$$iv) 3nR(150) = + 0.5(150 - \frac{300}{16}) nR$$

$$3nR(T_2 - T_1) = -P_{ext} \left(\frac{T_2}{P_2} - \frac{T_1}{P_1} \right)$$

$$\frac{7T_2}{2} = \frac{300}{2} + 900 \quad 3(T_2 - T_1) = -P_{ext} \left(\frac{T_2}{P_2} - \frac{T_1}{P_1} \right)$$

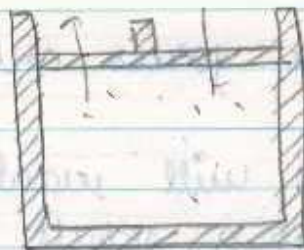
$$3T_2 - 3T_1 = -\frac{T_2}{2} + \frac{T_1}{22}$$

From FLT,

$$\Delta U = q + w$$

for adiabatic, $q = 0$

$$\Rightarrow \Delta U = w$$



If work is done by the system.

$$w = -20 \text{ kJ}$$

$$\Delta U = -20 \text{ kJ}$$

work is done at the expense of internal energy

If work is done on the system

$$w = 20 \text{ kJ}$$

$$\Delta U = 20 \text{ kJ}$$

increase in internal energy

if expansion is free

$$w = 0$$

$$\Delta U = 0$$

$$\Delta T = 0 \rightarrow \text{Ideal gas}$$

• Free expansion of real gas

From FLT,

$$\Delta U = q + w \Rightarrow 0$$

$$\Delta U = 0$$

$$\Delta U = KE + PE = 0$$

$$KE = f(T)$$

If attractive forces are dominating ($z < 1$),

gas molecules will want to shrink but gas is expanding, as a result, stability will decrease, means $PE \uparrow$, $KE \downarrow$ & $T \downarrow$. Gas is showing cooling effect.

If repulsive forces are dominating ($z > 1$),

gas molecules will try to expand due to repⁿ, so on free expansion stability will increase, means $PE \downarrow$, $KE \uparrow \Rightarrow T \uparrow$. Gas is showing heating effect.

• Most of the gases will show cooling effect on free expansion. (except H_2 & He)

• If Vanderwaal gas is mentioned then we have to consider only attractive forces.

• If free expansion takes place at inversion temp. then there will be no heating or cooling.

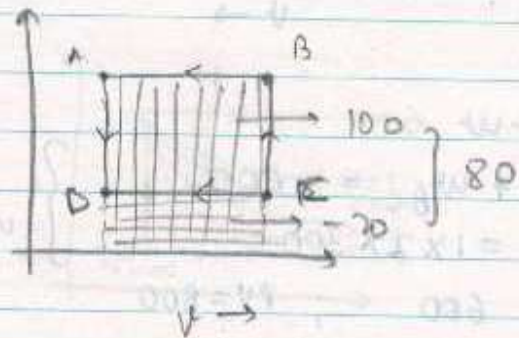
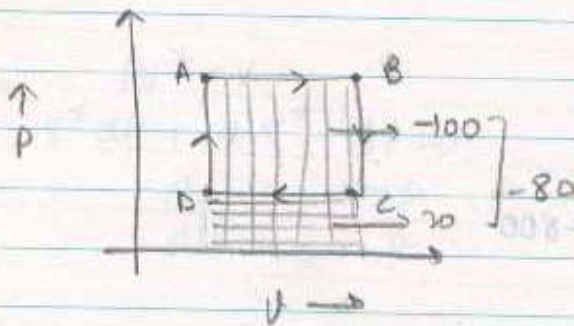
Attractive forces dominate (cooling effect) $\xleftarrow{T < T_i^*}$ $T_i^* = \frac{2a}{bR}$ $\xrightarrow{T > T_i^*}$ Repulsive forces dominate (heating effect)

$\downarrow$
force of attⁿ = force of repⁿ
($\Delta T = 0$)

* Cyclic Process $\rightarrow$

Sign convention

Anticlockwise work is taken as +ve & clockwise work is taken as -ve coz in anticlockwise work expansion occurs at low pressure & compression occurs at high pressure.



- $\Delta(SF) = 0 = \Delta H = \Delta U = \Delta G = \Delta S$

- From FLT,

$$\Delta U^0 = q + w$$

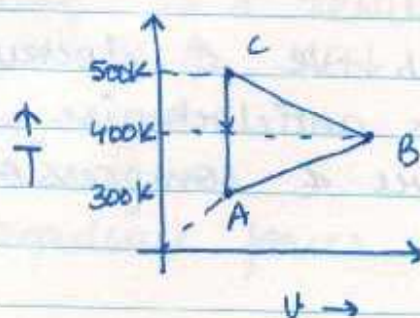
$$q = -w$$

$$w = w_{AB} + w_{BC} + w_{CD} + w_{DA}$$

Note $\rightarrow$ In case of standard process like isobaric, isochoric, isothermal we can directly use formula. But if process is not standard then

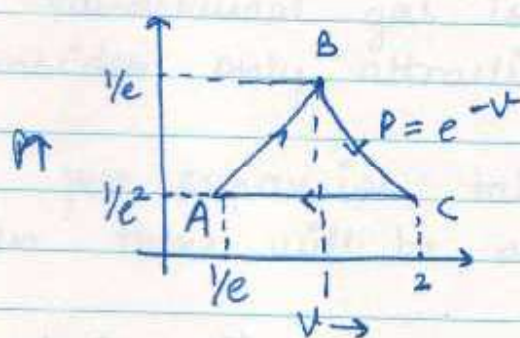
$$w = - \int_{V_1}^{V_2} P dV$$

8. Calculate work done in process BC for 1 mol of an ideal gas if total 600 cal heat is released by gas in whole process.



$$\begin{aligned} \rightarrow q &= -w \\ w_{AB} + w_{BC} &= -600 \\ P \times V &= 1 \times 2 \times 300 \\ PV &= 600 \quad ; \quad PV = 800 \end{aligned} \quad \left. \vphantom{\begin{aligned} w_{AB} + w_{BC} &= -600 \\ P \times V &= 1 \times 2 \times 300 \\ PV &= 600 \end{aligned}} \right\} w = 800$$

9. Calculate 'w'



$$\begin{aligned} \rightarrow w &= (2 - 1/e) \frac{1}{e^2} - \left[\frac{1}{2} \left(1 - \frac{1}{e} \right) \left(\frac{1}{e} - \frac{1}{e^2} \right) + \left(1 - \frac{1}{e} \right) \frac{1}{e^2} \right] - \left[\frac{1}{e^2} + \left[\frac{1}{e^2} - \frac{1}{e} \right] \right] \\ &= \frac{2}{e^2} - \frac{1}{e^3} - \frac{1}{2e} + \frac{1}{2e^3} - \frac{1}{e^2} - \frac{1}{e^2} + \frac{1}{e} \\ &= \frac{\pi \phi}{\theta} \left(\frac{1}{2e} - \frac{1}{2e^3} \right) \end{aligned}$$

* Polytropic Process $\Rightarrow$

It is a general process represented by $PV^x = k$

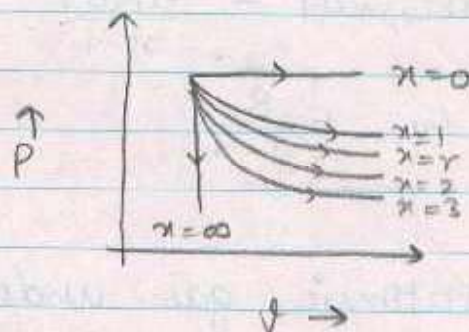
if $x=1 \Rightarrow PV = k \Rightarrow$ Isothermal

$x=0 \Rightarrow P = k \Rightarrow$ Isobaric

$x=\gamma \Rightarrow PV^\gamma = k \Rightarrow$ Adiabatic

$x=\infty \Rightarrow V = k \Rightarrow$ Isochoric

$$PV^x = k$$
$$V^x dP + x V^{x-1} dV = 0$$
$$\frac{dP}{dV} = -\frac{xP}{V}$$



• Work done :

$$W = - \int_{V_1}^{V_2} P dV$$
$$= - \int_{V_1}^{V_2} \frac{k}{V^x} dV$$

$$P_1 V_1^x = P_2 V_2^x = k$$

$$= - \frac{k(x-1)}{V^{x-1}} \Big|_{V_1}^{V_2}$$

$$= - \frac{k(x-1)}{V_2^{x-1}} + \frac{k(x-1)}{V_1^{x-1}}$$

$$= P_2 V_2 (1-x) + P_1 V_1 (x-1)$$

$$= \frac{nR\Delta T}{x-1}$$

Ex 1: till 48
Ex 2: 1-19
DPP

• Molar heat capacity for polytropic process:

$$C_m = \frac{q}{n \Delta T}$$

$$\Delta U = q + w$$

$$n C_{v,m} \Delta T = q + \frac{n R \Delta T}{\gamma - 1}$$

$$q = n C_{v,m} \Delta T - \frac{n R \Delta T}{\gamma - 1}$$

$$C_m = C_{v,m} + \frac{R}{1 - \gamma}$$

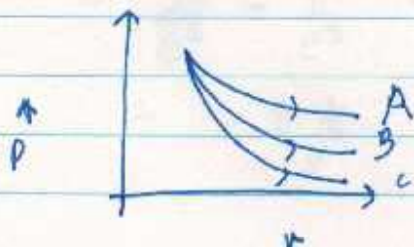
Q. for monoatomic gas undergoing process $P = kV$
find C_m .

$$\rightarrow C_m = \frac{3}{2}R + \frac{R}{2} = \underline{\underline{2R}}$$

Q. for monoatomic gas undergoing process $T = kV$
find C_m

$$\rightarrow C_m = \frac{3}{2}R + R = \underline{\underline{\frac{5R}{2}}}$$

Q. Three gases are taken at same initial state & under adiabatic reversible exp. Match them correctly



gases

A	→	H ₂ (g)
B	→	He (g)
C	→	SO ₂ (g)

• Graphical comparison b/w reversible & irreversible
Adiabatic process :-

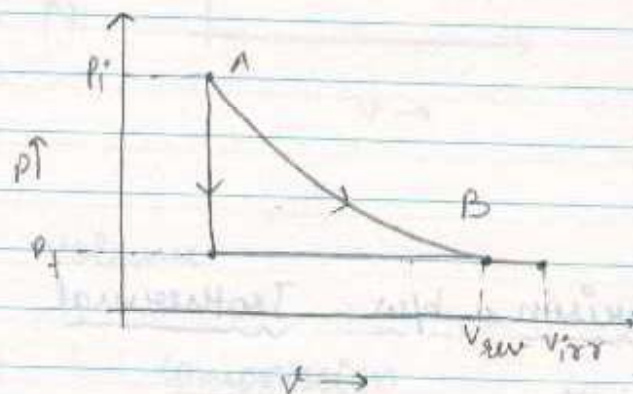
A] for same final pressure :

	U_i	w	U_f
rev	100	-60	40
irr	100	-40	60

$$(U_f)_{irr} > (U_f)_{rev}$$

$$T_{irr} > T_{rev}$$

$$\therefore V_{irr} > V_{rev}$$



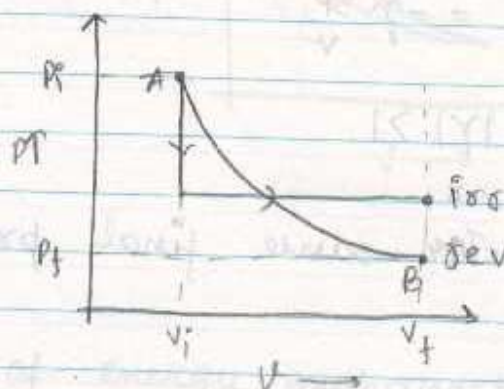
B] for same final volume :

	U_i	w	U_f
rev	100	-60	40
irr	100	-40	60

$$(U_f)_{irr} > (U_f)_{rev}$$

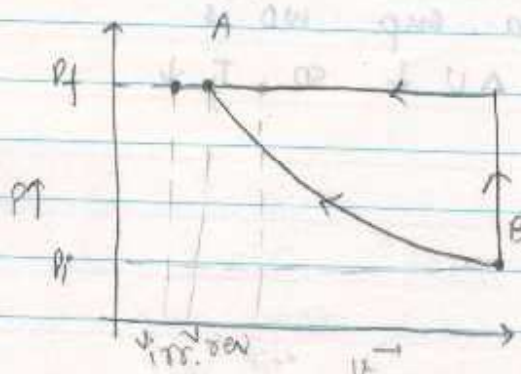
$$T_{irr} > T_{rev}$$

$$\therefore P_{irr} > P_{rev}$$

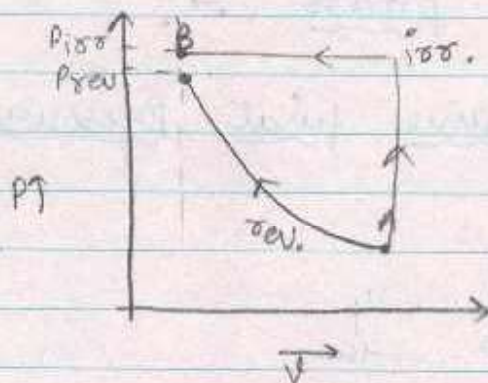


COMPRESSION :-

A] for same pressure :



B7 For same final volume.



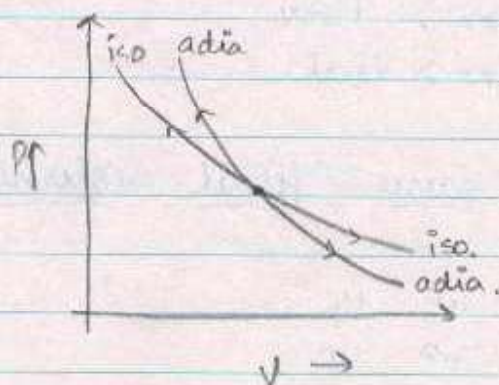
Comparison b/w Isothermal & Adiabatic Process $\Rightarrow$

$$PV^\gamma = K$$

$$V^\gamma dp + \gamma V^{\gamma-1} dV p = 0$$

$$\frac{dp}{dV} = -\gamma \frac{P}{V}$$

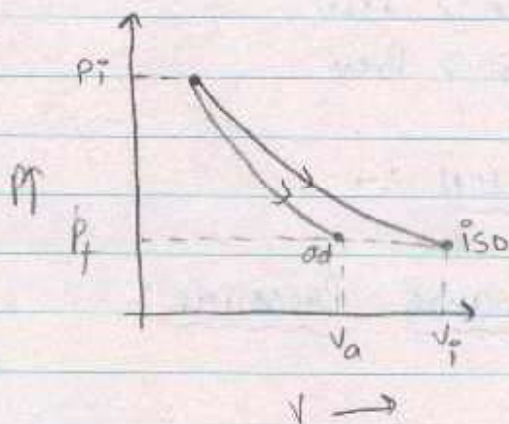
$$|\gamma| > 1.$$



Case I : For same final pressure.

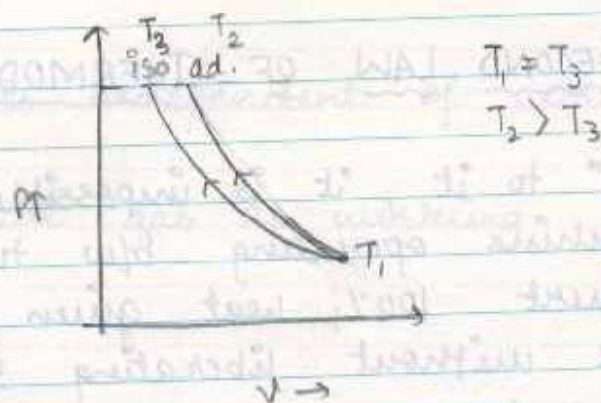
Expansion

As in isothermal process,
 $\Delta T = 0 \Rightarrow \Delta U = 0 \Rightarrow U = \text{const.}$
 But in adia. exp. w is
 -ve, hence $\Delta U \downarrow$ so $T_f \downarrow$
 $\therefore V_f \downarrow$.



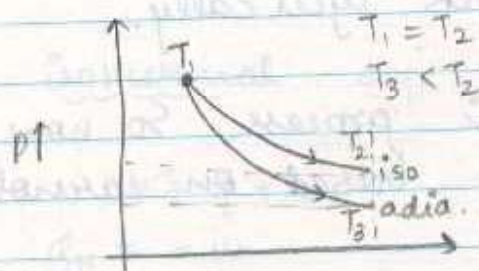
Compression

$V_{ad.} > V_{iso} \Rightarrow T_{ad.} > T_{iso}$
in case of compression,
 $w = +ve$ hence $\Delta U \uparrow$
 $\therefore T_f \uparrow$ (adia.) whereas
in isothermal $T_i = T_f$.

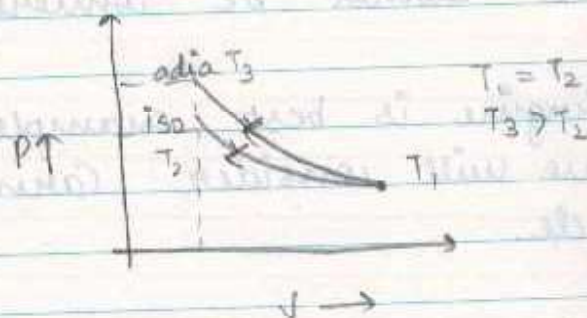


Case II: For same final volume

Expansion



Compression



* Limitations of FLT :->

- 1) FLT helps in determination of energy exchanges in terms of heat and work, but it does not tell anything about :
 - a) feasibility of a process
 - b) extent of a process

* SECOND LAW OF THERMODYNAMICS :

Accⁿ to it, it is impossible to build a cyclic machine operating b/w two temperatures which can convert 100% heat given to it at high temp. into work without liberating some amt of heat at low temperature.

OR

There is a natural asymmetry b/w heat & work. 100% work can be converted to heat but 100% ~~work~~ heat cannot be converted to work cyclically.

- Engine is best example of cyclic process. So now we will consider Carnot engine based on Carnot cycle.

* Carnot Cycle :

It consists of 4 reversible process, out of which ~~two~~ two are rev. isothermal (C & E) & two are rev. adiabatic (C & E).

$$\eta = \frac{\text{work done}}{\text{heat supplied}} = \frac{-w}{Q_H} = \frac{|w|}{Q_H} = \frac{Q_H + Q_L}{Q_H}$$

where,

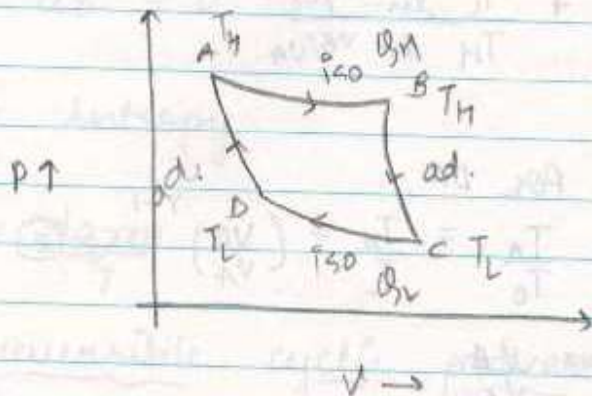
Q_H → Heat supplied to system

Q_L → Heat rejected at sink

w → work done per cycle.

- Efficiency of Carnot cycle is independent of working substance.

But we will consider ideal gas as working substance.



For isothermal exp. (AB):

$$\Delta U = q + w$$

$$Q_H = -w$$

$$= nRT_H \ln \left(\frac{V_B}{V_A} \right)$$

$$W_{AB} = -nRT_H \ln \left(\frac{V_B}{V_A} \right)$$

For adiabatic exp. (BC):

$$\Delta U = q + w$$

$$W_{BC} = nC_{v,m} (T_L - T_H)$$

For isothermal comp. (CD):

$$Q_L = -w = -(-nRT_L \ln \left(\frac{V_D}{V_C} \right))$$

2:49-51

2:20-21, 38, 43, 49,
53, 55-58for adiabatic comp. (DA):

$$W_{DA} = \Delta U = nC_{v,m}(T_H - T_L)$$

$$\eta = \frac{Q_H + Q_L}{Q_H} = 1 + \frac{Q_L}{Q_H} = 1 + \frac{T_L \ln V_D/V_C}{T_H \ln V_B/V_A}$$

for BC,

$$\frac{T_B}{T_C} = \frac{T_H}{T_L} = \left(\frac{V_C}{V_B}\right)^{\gamma-1} \quad \text{--- (1)}$$

for DA,

$$\frac{T_A}{T_D} = \frac{T_H}{T_L} = \left(\frac{V_D}{V_A}\right)^{\gamma-1} \quad \text{--- (2)}$$

$$\frac{V_C}{V_B} = \frac{V_D}{V_A} = \frac{V_D}{V_C} = \frac{V_A}{V_B}$$

$$\boxed{\eta_{\text{carnot}} = 1 - \frac{T_L}{T_H}}$$

for ideal gas & carnot cycle

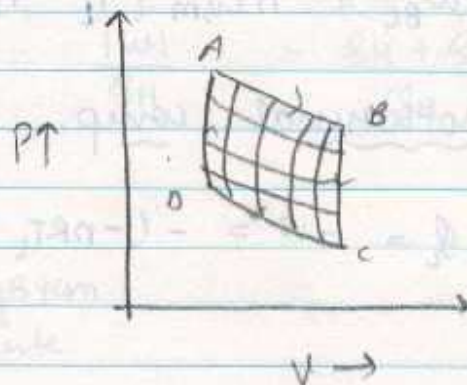
In order to have 100% efficiency either T_L should be 0K or $T_H \rightarrow \infty$ & practically both conditions are impossible so we can say 100% efficiency is not possible, in case of carnot cycle.

* for reversible cyclic process:

$$1 - \frac{T_L}{T_H} = 1 + \frac{Q_L}{Q_H}$$

$$\Rightarrow -\frac{Q_H}{T_H} = \frac{Q_L}{T_L}$$

$$\frac{(Q_L)_{\text{rev}}}{T_L} + \frac{(Q_H)_{\text{rev}}}{T_H} = 0$$



If cycle is divided into n parts

$$\oint \frac{dq_{rev}}{T} = 0$$

$$\oint ds = 0$$

$ds \rightarrow$ entropy

$$ds = \frac{dq_{rev}}{T}$$

• For irreversible cyclic process:

$$(\eta_{rev}) > (\eta_{irr})$$

$$1 - \frac{T_L}{T_H} > 1 + \frac{(Q_L)_{irr}}{(Q_H)_{irr}}$$

$$\Rightarrow \frac{(Q_L)_{irr}}{T_L} + \frac{(Q_H)_{irr}}{T_H} < 0$$

For n steps

$$\oint \frac{dq_{irr}}{T} < 0$$

In general,

$$\oint \frac{dq}{T} \leq 0$$

* Claussius Inequality :

• Non cyclic process :

Suppose we can move from A to B by two paths one is reversible & another is irreversible.



$$dU_{rev} = dq_{rev} + dw_{rev} \quad \text{--- (1)}$$

$$dU_{irr} = dq_{irr} + dw_{irr} \quad \text{--- (2)}$$

$\therefore U$ is a state f.n.,

$$dq_{rev} + dw_{rev} = dq_{irr} + dw_{irr}$$

$$\Rightarrow dq_{rev} - dq_{irr} = -P_{ext} dv + P_g du \quad [\because \text{for rev } P_{ext} = P_g]$$

$$\Rightarrow dq_{rev} - dq_{irr} = (P_g - P_{ext}) dv$$

Case I : if $P_{ext} > P_{gas} \Rightarrow$ compression

$$dq_{rev} - dq_{irr} = \underline{(P_g - P_{ext})} \underline{dv}$$

$$dq_{rev} - dq_{irr} > 0$$

Case II: if $P_g > P_{ext} \Rightarrow$ Expansion

$$dq_{rev} - dq_{irr} > 0$$

$$Tds > dq_{irr}$$

$$\boxed{ds > \frac{dq_{irr}}{T}} \rightarrow \text{non cyclic}$$

In general, $ds \geq \frac{dq}{T}$

If system is isolated,

$$\begin{aligned} dq &= 0 \\ ds &> 0 \end{aligned}$$

* Introduction to Entropy $\Rightarrow$

$$ds = \frac{dq_{rev}}{T}$$

$$\int ds = \int \frac{dq_{rev}}{T}$$

$$\Delta S = \int \frac{dq_{rev}}{T} \quad \text{JK}^{-1}$$

$S \rightarrow$ State f:n

* Calculation of ΔS :

• Case I : For rev. heating / cooling of solids / liquids

$$dS = dq_{\text{rev}}$$

$$|dS = n \int C_m \frac{dT}{T}$$

If C_m is independent of temp.

$$\Delta S = n C_m \ln \frac{T_2}{T_1}$$

Case II : In case of real gas undergoing isobaric rev. process

$$dS = \left(\frac{dq_{\text{rev}}}{T} \right)_P = n C_{p,m} \frac{dT}{T}$$

$$\Delta S = n C_{p,m} \ln \frac{T_2}{T_1}$$

Case III : In case of real gas undergoing isochoric rev. process

$$dS = \left(\frac{dq_{\text{rev}}}{T} \right)_V = n C_{v,m} \frac{dT}{T}$$

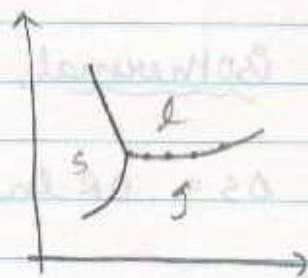
$$\Delta S = n C_{v,m} \ln \frac{T_2}{T_1}$$

Case IV : If phase transformation occurs at BP, MP or sublimation point, then process is considered as rev.



$$\Delta S_{\text{vap}} = \frac{q_{\text{rev}}}{T} = \frac{\Delta H_{\text{vap.}}}{T_{\text{BP}}}$$

$$\Delta S_{\text{fus.}} = \frac{\Delta H_{\text{fus}}}{T_{\text{MP}}}$$



Case V : For an ideal gas undergoing any process.

$$\Delta U = q + w$$

$$\Delta U - w = q$$

$$\frac{1}{T} n R \Delta T - P \Delta V$$

$$\Delta S = \int \frac{dq_{\text{rev}}}{T}$$

$$\Rightarrow \Delta S = \int \frac{du - dw}{T} = \int \frac{n C_{v,m} dT}{T} - \int \frac{P dv}{T}$$

$$= n C_{v,m} \ln \frac{T_2}{T_1} + n R \ln \frac{V_2}{V_1}$$

$$\Delta S = n C_{v,m} \ln \left(\frac{P_2 V_2}{P_1 V_1} \right) + n R \ln \frac{V_2}{V_1}$$

$$\Delta S = n C_{v,m} \ln \left(\frac{T_2}{T_1} \right) + n R \ln \frac{P_1}{P_2}$$

For isobaric,

$$\Delta S = n C_{v,m} \ln \left(\frac{T_2}{T_1} \right)$$

• For isochoric,

$$\Delta S = n C_{v,m} \ln \frac{T_2}{T_1}$$

For isothermal,

$$\Delta S = nR \ln \frac{V_2}{V_1}$$

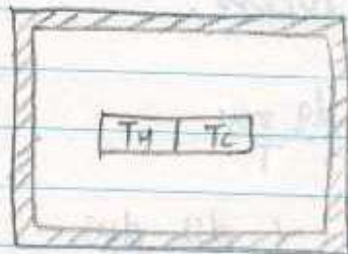
* Using entropy to find out direction $\Rightarrow$

Suppose we have taken an adiabatic rigid vessel & inside it we have taken two rods at diff. temperatures.

where,

T_H & $T_C \rightarrow$ temp. of hot & cold rods respec.

Q_H , $Q_C \rightarrow$ heat exchanged by hot & cold rod resp.



$$Q_H + Q_C = 0$$

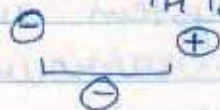
$$\Delta S_T = \Delta S_H + \Delta S_C$$

$$\Delta S_T = \frac{-Q_C}{T_H} + \frac{Q_C}{T_C}$$

$$\Delta S_T = Q_C \left(\frac{T_H - T_C}{T_H T_C} \right)$$

Case I : if heat flows from cold to hot rod

$$\Delta S = Q_c \left(\frac{T_H - T_c}{T_H T_c} \right)$$



$$\Delta S_T < 0 \Rightarrow S_f < S_i$$

Case II : if heat flows from hot to cold

$$\Delta S_T = Q_c \left(\frac{T_H - T_c}{T_H T_c} \right)$$

$$\Delta S > 0$$

$$\Rightarrow S_f > S_i$$

For an isolated system,

if $\Delta S > 0 \Rightarrow$ process is spontaneous

if $\Delta S < 0 \Rightarrow$ process is non-spontaneous

if $\Delta S = 0 \Rightarrow$ process is rev. or at equilibrium

Q. True / False

1) If $\Delta S > 0$ process must be spontaneous (F)

2) If $\Delta S < 0$ in an isolated system then process must be non spontaneous. (T)

Conclusion : In an isolated system increase in entropy represents direction of natural change.

generally we deal with closed system and not with isolated coz system interacts with surrounding.

System + Surrounding = Universe

$$\Delta S_{\text{uni/total}} = \Delta S_{\text{sys}} + \Delta S_{\text{sur}}.$$

if $\Delta S_{\text{Total}} > 0 \Rightarrow$ process is spontaneous

$\Delta S_{\text{Total}} < 0 \Rightarrow$ process is non-spontaneous

$\Delta S_{\text{Total}} = 0 \Rightarrow$ " " reversible or at equilibrium

Q. True / False

1) If $\Delta S < 0 \Rightarrow$ process must be non spontaneous. F

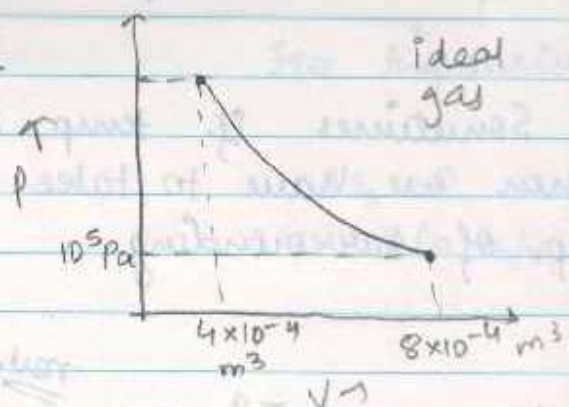
2) If $\Delta S_{\text{uni}} > 0 \Rightarrow$ process must be spontaneous. T

All natural processes are spontaneous and irreversible. It means entropy of universe is increasing continuously. From above points it is clear that entropy of universe can be created but it cannot be destroyed while energy can neither be created nor be destroyed.

- Q. For an isothermal process at 300K calculate ΔS
- if process is reversible
 - if process is irreversible

$$\rightarrow P_1 = 10^5 \times 2 \text{ Pa} ; V_1 = 4 \times 10^{-4} \text{ m}^3$$

$$P_2 = 10^5 \text{ Pa} ; V_2 = 8 \times 10^{-4} \text{ m}^3$$



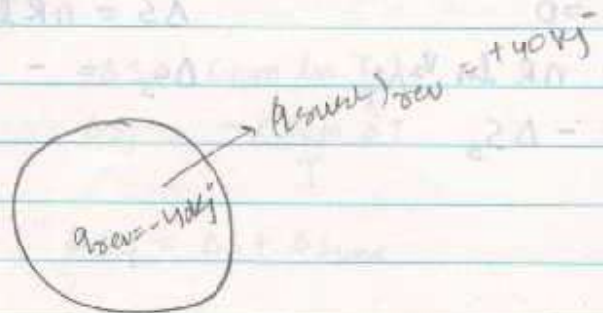
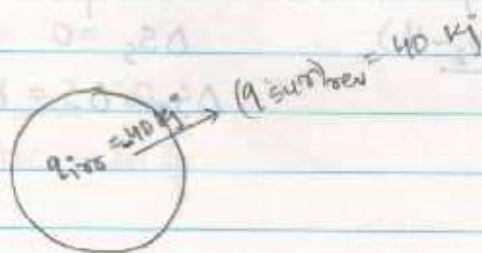
$$\text{i) } \Delta S = nR \ln \frac{V_2}{V_1} = \frac{80}{300} \ln 2$$

$$\text{ii) } \Delta S = -nR \ln \frac{P_2}{P_1} = \frac{80}{300} \ln 2$$

In irreversible isothermal process, heat exchange will be irr. & we know that $\Delta S \neq \frac{q_{irr}}{T}$. In such a scenario we will connect initial state to final state thru a reversible path and calculate ΔS for that. In irreversible we will use q_{rev} for path we have assumed so q_{rev} is not actual heat exchange.

• Calculation of ΔS_{sur} :

Surrounding is always considered as infinitely large and and each b/w system & surr cannot change pressure & temp. of surr. as a result P & T of surr will remain constant.



$$\Delta S_{\text{surround.}} = \frac{q_{\text{surround.}}}{T} = -\frac{q_{\text{sys}}}{T}$$

Note: Sometimes if temp. of surr. is not mentioned then we have to take final temp. of system as temp. of surrounding.

Q.

$$\begin{aligned} \rightarrow \Delta S_{\text{surround.}} &= -\frac{q_{\text{sys}}}{T} \quad \text{rev.} \\ &= -\int \frac{dV}{\frac{V}{T}} = -\frac{80 \ln 2}{300} \end{aligned}$$

$$\begin{aligned} \Delta S_{\text{Total}} &= \Delta S_{\text{surround.}} + \Delta S_{\text{sys}} \\ &= \frac{160}{300} \ln 2 = 0. \end{aligned}$$

$$\begin{aligned} \Delta S_{\text{surround.}} &= -\frac{q_{\text{sys}}}{T} \quad \text{irr.} \\ &= -\frac{10^5 (4 \times 10^{-4})}{300} = -\frac{40}{300} \end{aligned}$$

$$\Delta S_T = \frac{80}{300} \ln 2 - \frac{40}{300}$$

• Calculation of ΔS for various processes:

Isothermal

Reversible

$$\Delta S_T = \Delta S + \Delta S_S$$

$$\Delta S_T = 0$$

$$\Delta S = nR \ln V_2/V_1$$

$$\Delta S = -\Delta S_S$$

Irreversible

$$\Delta S_T = \Delta S + \Delta S_S$$

$$\Delta S = nR \ln V_2/V_1$$

$$\Delta S_S = -\frac{P_{\text{ext}}(V_2 - V_1)}{T}$$

Free Expansion

It will also be adiabatic

$$\Delta S_S = 0$$

$$\Delta S_T = \Delta S = nR \ln V_2/V_1$$

Adiabatic

Reversible

$$\Delta S_T = \Delta S + \Delta S_s$$

$$\Delta S_s = 0$$

$$\Delta S_T = 0$$

$$\Delta S = 0$$

Irreversible

$$\Delta S_s = 0$$

$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

Free Expansion

$$\Delta S_s = 0$$

$$\Delta S = nR \ln \frac{V_2}{V_1}$$

g. Prove that for reversible adiabatic process $\Delta S = 0$.

→ As process is adiabatic,

$$q_{\text{sys}} = 0 \Rightarrow \Delta S_{\text{sur}} = 0$$

ΔS_T is also 0 as process is reversible.

$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$TV^{r-1} = K \Rightarrow V^{r-1} = \frac{K}{T} \Rightarrow V_1 = \left(\frac{K}{T_1}\right)^{\frac{1}{r-1}}; V_2 = \left(\frac{K}{T_2}\right)^{\frac{1}{r-1}}$$

$$\frac{V_2}{V_1} = \left(\frac{T_1}{T_2}\right)^{r-1} \Rightarrow \Delta S = nC_{v,m} \ln \frac{T_2}{T_1} + (r-1)nR \ln \frac{T_2}{T_1}$$

Isochoric

Reversible

$$\Delta S_T = 0$$

$$\Delta S + \Delta S_{\text{sur}} = 0$$

$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1}$$

Irreversible

$$\Delta S = nC_{v,m} \ln \frac{T_2}{T_1}$$

$$\Delta S_s = -\frac{nC_{v,m} \Delta T}{T}$$

$$\Delta S_T = \Delta S + \Delta S_{\text{sur}}$$

Isobaric

Reversible

Irreversible

Free Expansion

$$\begin{aligned}\Delta S_T &= 0 \\ -\Delta S &= \Delta S_{\text{surroundings}} \\ &= -n C_{p,m} \ln \frac{T_2}{T_1}\end{aligned}$$

$$\Delta S = n C_{p,m} \ln \frac{T_2}{T_1}$$

X

$$\begin{aligned}\Delta S_{\text{surroundings}} &= -\frac{[n C_{v,m} \Delta T - P \Delta V]}{T} \\ &= -\frac{n C_{p,m} \Delta T}{T}\end{aligned}$$

Q. 1 mol of a non linear triatomic ideal gas is expanded adiabatically at 300K from 16 atm to 1 atm. Find ΔS , $\Delta S_{\text{surroundings}}$ & ΔS_T

- i) if exp. is carried out reversible
- ii) if " " " " " irr.
- iii) free exp.

$$\rightarrow \text{i) } \Delta S = \Delta S_S = \Delta S_T = 0$$

$$\text{ii) } \Delta S_S = 0$$

$$\Delta S = n C_{v,m} \ln \frac{T_2}{T_1}$$

$$= 1 \times 3R \ln \frac{229.68}{300} + R \ln \frac{1}{16}$$

$$\text{iii) } \Delta S_S = 0$$

$$\Delta S = nR \ln V_2/V_1 = \Delta S_T$$

$$= R \ln V_2/V_1 = R \ln 1/16$$

$$P^{1/\gamma} = \frac{K}{V}^{1/\gamma}$$

$$\gamma = 1 + \frac{2}{f}$$

$$PV^\gamma = K \Rightarrow T P^{1-\gamma} = K$$

$$TV^{\gamma-1} = K$$

$$T^{1/\gamma} = \frac{K}{V}^{1/\gamma}$$

$$T P^{1-\gamma} = K$$

$$P^{1/\gamma}$$

$$3R(T_2 - T_1) = -P \left(\frac{RT_2}{P_2} - \frac{RT_1}{P_1} \right)$$

$$3T_2 - 3T_1 = \frac{300}{16} - T_2$$

$$4T_2 = \frac{300}{16} + \frac{225}{4} = \frac{300}{16} + \frac{900}{16}$$

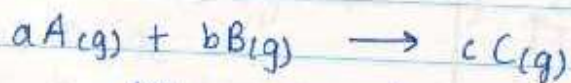
$$T_2 = 229.68$$

* Physical significance of Entropy:

- It represents extent of randomness.
 $S_g \gg S_l > S_s$
- On $\uparrow$ temp., randomness will increase.

Sl. No.	Process	ΔS
1.	Sublimation	+ve
2.	Vaporisation	+ve
3.	Melting	+ve
4.	Melting of ice	+ve
5.	Boiling of egg	+ve {denatured protein}

- During chemical rxn,



$$\Delta n_g = c - a - b$$

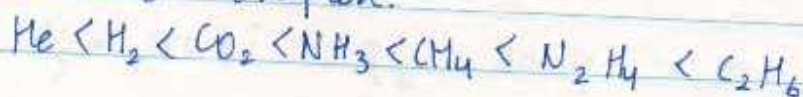
if,

$$\Delta n_g > 0 \Rightarrow \Delta S = +ve \Rightarrow \text{entropy will } \uparrow$$

$$\Delta n_g < 0 \Rightarrow \Delta S = -ve \Rightarrow \text{entropy will } \downarrow$$

$$\Delta n_g = 0 \Rightarrow \Delta S \text{ can be } +ve \text{ or } -ve.$$

- As atomicity $\uparrow$, randomness or entropy $\uparrow$ as molecule becomes complex.



1B $\rightarrow$ A, D, A, A, ABC, BD, BC, ABCD, 280, 27.7

17 $\rightarrow$ A, A, C, A, CD, C, A, A

Sheet $\Rightarrow$ 3(A) 1-49 (4x8x10x)
3(B) complete

RB $\Rightarrow$ 9, 10, 11, 21, 22,
36, 30-38

i) $n \left[\frac{Q}{T} \right]$

ii) $v \left[\frac{Q}{T} \right]$

- If atomicity is same then entropy $\uparrow$ with molar mass.



{ internal exchange of e^- }



{ $O_2(g) \rightarrow 2O(g)$ $\Delta_{\text{orig}} = +ve$ }

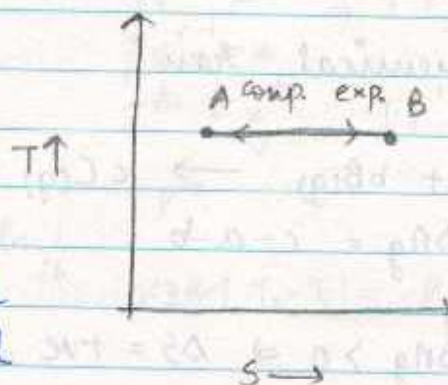


* T-S diagram $\Rightarrow$

1) For reversible isothermal process.

$$ds = \frac{dq_{\text{rev}}}{T}$$

$$dq_{\text{rev}} = T ds$$



Area of T-S curve will give magnitude of heat exchanged during reversible isothermal process.

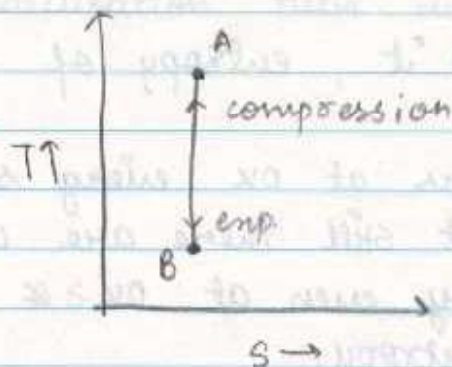
2] For reversible adiabatic process :

$$ds = \frac{dq_{rev}}{T}$$

For adiabatic process,

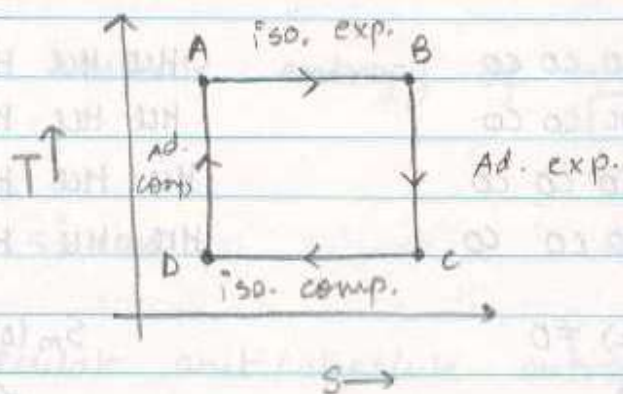
$$dq = 0$$

$$\therefore ds = 0$$



3] For Carnot cycle :

Area of this diagram will give net heat exchanged or net work done during the cycle.



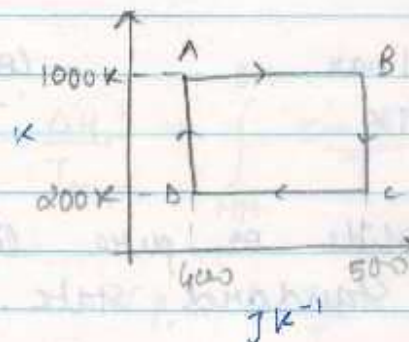
Q. From T-S diagram of Carnot cycle calculate

- η
- work done per cycle
- heat taken from the source & rejected at sink.

→ a) $\eta = 4/5$

b) $-q_{net} = 800 \times 100 \text{ J}$

c) $q_s = 100 \text{ KJ}$ $q_{si} = 20 \text{ KJ}$



* Third law of Thermodynamics :-

Accⁿ to it, entropy of substance at 0K will be zero.

It means at 0K every substance will be perfect crystal. But still there are certain substances which have some entropy even at 0K & that entropy is called residual entropy.

eg,

CO & NO

HCl

CO CO CO CO CO
CO CO OC CO CO
CO CO CO CO CO
CO CO CO CO CO

$S_m(0K) \neq 0$
 ∞

HCl HCl HCl HCl HCl
HCl HCl HCl HCl HCl
HCl HCl HCl HCl HCl
HCl HCl HCl HCl HCl

$S_m(0K) = 0$
 ∞

• Standard state :

For gases

For solution (liquid)

$P = 1 \text{ bar}$
 $T = T_K$

$\text{conc.} = 1 \text{ M}$
 $T = T_K$

Pure solids or pure liquids are considered to be in their standard state.

Note : Sometimes at standard state, temp. is not given & we need temp. for calculation then we will use $T = 298\text{K}$.

Standard enthalpy change = ΔH°

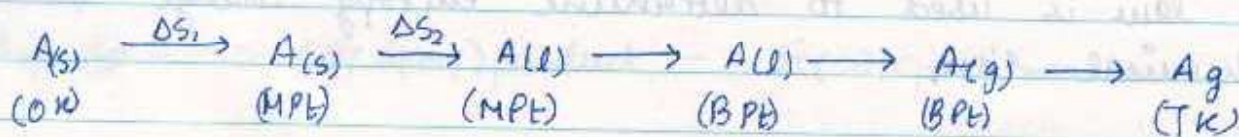
Standard entropy change = ΔS°

• Application of Third law of thermodynamics :

- 1) It is used to calculate absolute entropy of a substance.

$$\Delta S_{TK}^\circ = S_m^\circ(TK) - S_m^\circ(OK)$$

Suppose we want to calculate amt/absolute entropy of $A(g)$ at TK



For $n=1$ mol,

$$\begin{aligned} \Delta S_m^\circ = S_m^\circ(TK) - S_m^\circ(OK) &= \int_0^{MPE} \frac{C_m dT}{T} + \frac{\Delta H_{f.}}{T_{MPE}} + \int_{MPE}^{BPE} \frac{C_m dT}{T} \\ &+ \frac{\Delta H_{v.}}{T_{BPE}} + C_{p,m} \ln \frac{T_2}{T} \end{aligned}$$

* Third law of Thermodynamics :-

Accⁿ to it, entropy of substance at 0K will be zero.

It means at 0K every substance will be perfect crystal. But still there are certain substances which have some entropy even at 0K & that entropy is called residual entropy.

eg,

CO & NO

HCl

CO CO CO CO CO
CO CO OC CO CO
CO CO CO CO CO
CO CO CO CO CO

HCl HCl HCl HCl HCl
HCl HCl HCl HCl HCl
HCl HCl HCl HCl HCl
HCl HCl HCl HCl HCl

$$S_m(0K) \neq 0$$

$$S_m(0K) = 0$$

• Standard state:

for gases

for solution (liquid)

$$P = 1 \text{ bar}$$

$$T = T_K$$

$$\text{conc.} = 1 \text{ M}$$

$$T = T_K$$

Pure solids or pure liquids are considered to be in their standard state.

Note : Sometimes at standard state, temp. is not given & we need temp. for calculation then we will use $T = 298\text{K}$.

Standard enthalpy change = ΔH°

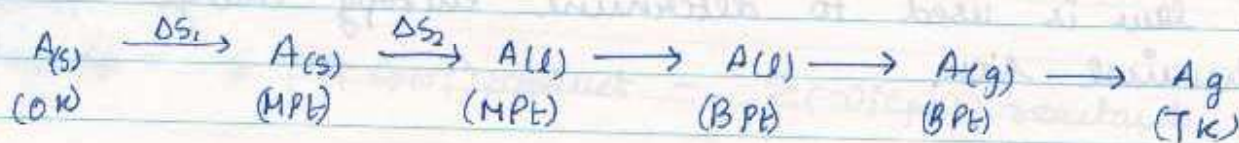
Standard entropy change = ΔS°

• Application of Third law of thermodynamics :

1) It is used to calculate absolute entropy of a substance.

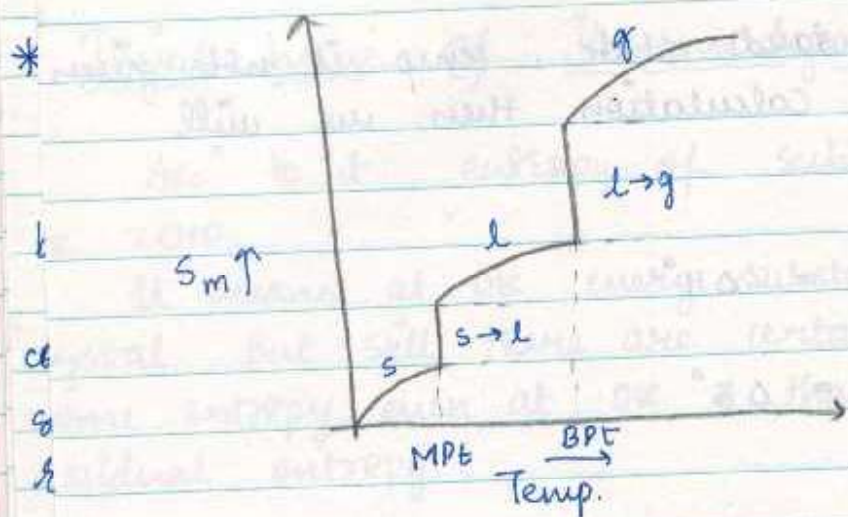
$$\Delta S_{TK}^\circ = S_m^\circ(TK) - S_m^\circ(OK)$$

Suppose we want to calculate amt/absolute entropy of $A(g)$ at TK



For $n=1$ mol,

$$\begin{aligned} \Delta S_m^\circ = S_m^\circ(TK) - S_m^\circ(OK) &= \int_0^{MP} \frac{C_m dT}{T} + \frac{\Delta H_f}{T_{m.p}} + \int_{MP}^{BP} \frac{C_m dT}{T} \\ &+ \frac{\Delta H_v}{T_{B.p}} + C_{p,m} \ln \frac{T_2}{T} \end{aligned}$$

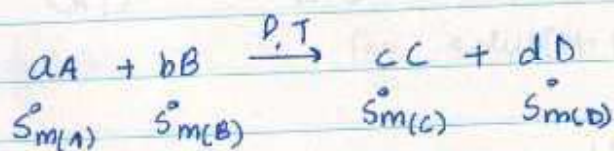


Note : Absolute entropy of all substances will be positive above 0K.

- $S_m^{\circ}(H^+) = 0$

In case of ions, molar entropy of H^+ at S.S. is taken as zero & entropies of other ions are calculated relative to it. As a result they can have +ve or -ve values.

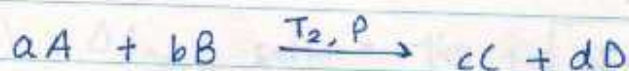
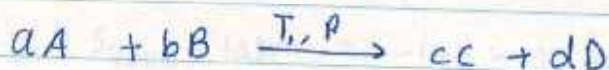
2) This law is used to determine entropy change during a chemical rxn.



$$\Delta_{\text{rxn}} S = (c S_m^{\circ}(C) + d S_m^{\circ}(D)) - (a S_m^{\circ}(A) + b S_m^{\circ}(B))$$

$$P \quad \Delta_{\text{rxn}} S = \sum (\nu_i S_{mi})_{\text{product}} - \sum (\nu_i S_{mi})_{\text{reactants}}$$

Case I : Variation of $\Delta_{\text{ran}} S$ with temp. at constant pressure



For 1 mol of rxn

$$dS = \frac{dq_{\text{rev}}}{T} = n C_{p,m} dT$$

$$\int_{\Delta S_{T_1}}^{\Delta S_{T_2}} d(\Delta S) = \Delta_{\text{ran}} C_p \int_{T_1}^{T_2} \frac{dT}{T}$$

$$\Delta S_{T_2} - \Delta S_{T_1} = \Delta_{\text{ran}} C_p \ln \frac{T_2}{T_1}$$

$$\Rightarrow \Delta S_{T_2} = \Delta S_{T_1} + \Delta_{\text{ran}} C_p \ln \frac{T_2}{T_1}$$

where,

$$\Delta_{\text{ran}} C_p = \sum (\nu_i C_{p,m,i})_{\text{product}} - \sum (\nu_i C_{p,m,i})_{\text{reactants}}$$

Case II : variation of $\Delta_{\text{ran}} S$ with temp. at constant volume

For 1 mol of rxn,

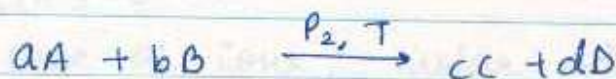
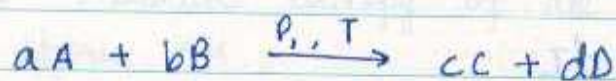
$$dS = \frac{dq_{\text{rev}}}{T} = n C_{v,m} dT$$

$$* \int_{\Delta S_{T_1}}^{\Delta S_{T_2}} d(\Delta S) = \Delta_{\text{ran}} C_{PV} \int_{T_1}^{T_2} \frac{dT}{T}$$

$$\Delta S_{T_2} - \Delta S_{T_1} = \Delta_{\text{ran}} C_{PV} \ln \frac{T_2}{T_1}$$

$$\Delta S_{T_2} = \Delta S_{T_1} + \Delta_{\text{ran}} C_{PV} \ln \frac{T_2}{T_1}$$

Case III : Variation of $\Delta_{\text{ran}} S$ with pressure at constant temp.



$$ds = \frac{dq_{\text{rev}}}{T} = \frac{\int P dv}{T} = -\frac{v dp}{T}$$

$$ds = -\frac{nR}{P} dp \Rightarrow \int_{\Delta S_{P_1}}^{\Delta S_{P_2}} d(\Delta S) = -\Delta n R \int_{P_1}^{P_2} \frac{dp}{P}$$

$$\Rightarrow \Delta S_{P_2} = \Delta S_{P_1} - \Delta n R \ln \frac{P_2}{P_1}$$

Q. For the rxn



Given at 1 atm, 300 K

$$S_{N_2} = 180 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$S_{O_2} = 220$$

$$S_{NO_2} = 240$$

$$C_{p,m}(N_2) = 30$$

$$C_{p,m}(O_2) = 30$$

$$C_{p,m}(NO_2) = 40$$

} -10

Calculate

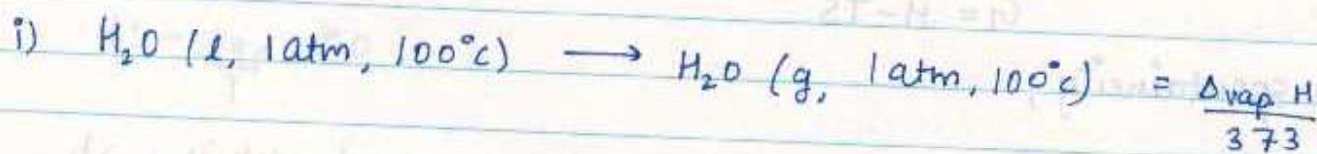
$$i) \Delta S_{300K, 1atm} = -140$$

$$ii) \Delta S_{400K, 1atm} = -140 - 10 \ln \frac{4}{3} = -143$$

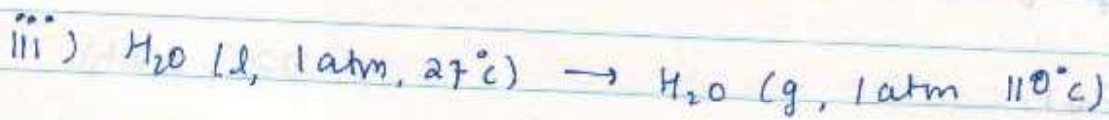
$$iii) \Delta S_{300K, satm} = -140 + R \ln 5$$

$$iv) \Delta S_{400K, satm} = -143 + R \ln 5$$

8. Calculate ΔS for 1 mol H_2O



$$\frac{40000}{373} + R \ln \frac{1}{2}$$



$$75 \times \ln \frac{373}{300} + \frac{40000}{373} + 35 \ln \frac{383}{373}$$

* Gibbs Free Energy (G):

- G is extensive qty. and state f.n.

Till now to predict spontaneity we have to do calculation for system as well as surrounding. If we want to find out spontaneity without doing any calculation for surrounding then we need a thermodynamic parameter for system which is G .

$$G = H - TS$$

For spontaneity,

$$dS_{\text{Total}} \geq 0$$

$$dS_{\text{sys}} + dS_{\text{sur}} \geq 0$$

$$dS_{\text{sys}} - \frac{dq_{\text{sys}}}{T} \geq 0$$

At constant P & T

$$dS - \frac{dq_p}{T} \geq 0$$

$$dq_p = dH \quad (\text{when } w_{\text{nonpv}} = 0)$$

$$dS - \frac{dH}{T} \geq 0$$

$$T dS - dH \geq 0$$

$$d(H - TS) \leq 0 \quad \Rightarrow \quad (dG)_{P,T} \leq 0 \quad (\text{when non})$$

DPP 19 $\Rightarrow$ C, B, B, C, B, D, ABCD, ABD, AD, AC, & PSR

$$ds_{\text{Total}} \geq 0$$

$$ds_{\text{sys}} - \frac{dq_{\text{sys}}}{T} \geq 0$$

At constant P & T

$$H = U + PV$$

$$dH = du + PdV + VdP$$

$$dH = dq_p + dw_{\text{non pv}}$$

$$dq_p = dH - dw_{\text{non pv}}$$

$$ds - \frac{dq_p}{T} \geq 0$$

$$ds - \left(\frac{dH - dw_{\text{non pv}}}{T} \right) \geq 0$$

$$(dG)_{P,T} \leq dw_{\text{non pv}}$$

If non pv work is zero

$$(dG)_{P,T} \leq 0$$

Q. True / False

- 1) If $\Delta G < 0 \Rightarrow$ Process must be spontaneous (F).
(Only at constant P & T).

→ * Relationship b/w ΔS_{Total} , $\Delta G_{P,T}$ & Temperature :

$$\Delta S_{Total} = \Delta S + \Delta S_{surv.}$$

$$\Delta S_{Total} = \Delta S - \frac{q_p}{T}$$

At constant P & T

$$\Delta S_{Total} = \Delta S - \frac{q_p}{T}$$

$$\Delta S_{Total} = \Delta S - \frac{\Delta H}{T}$$

$$T \Delta S_{Total} = -(\Delta H - T \Delta S)$$

$$\therefore \boxed{T \Delta S_T = -\Delta G_{P,T}} \rightarrow \text{When non PV work} = 0$$

closed system

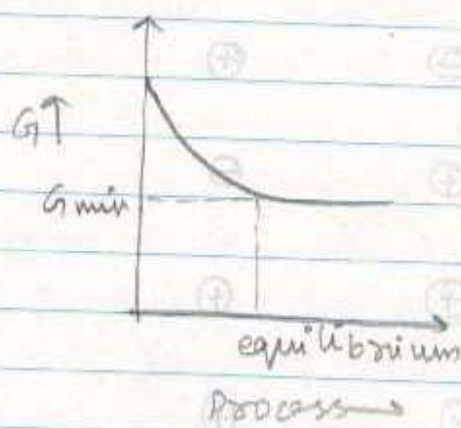
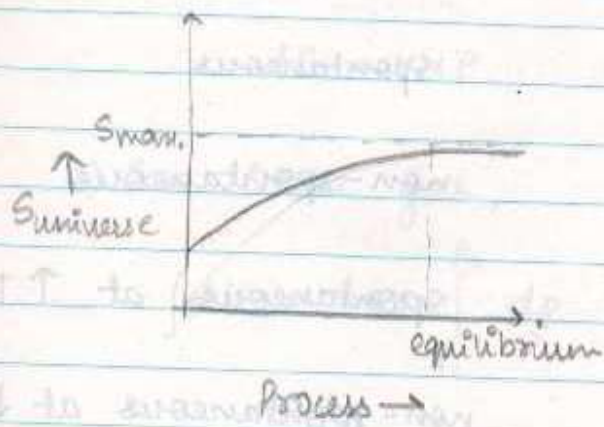
if $\Delta S_T > 0 \Rightarrow \Delta G_{P,T} < 0 \Rightarrow$ Process is spontaneous

if $\Delta S_T < 0 \Rightarrow \Delta G_{P,T} > 0 \Rightarrow$ Process is non spontaneous

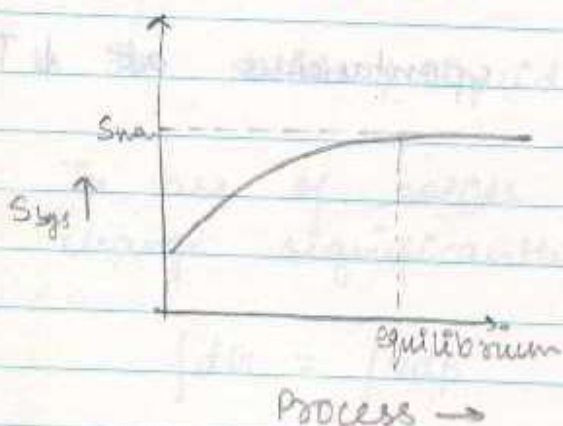
if $\Delta S_T = 0 \Rightarrow \Delta G_{P,T} = 0 \Rightarrow$ Process is reversible or at equilibrium

Still ΔS_T is ultimate criterion of spontaneity & G is defined for system.

For closed system,



For isolated system,



* Predicting spontaneity at constant P & T for chemical
 Ran $\rightarrow$

$$R \xrightarrow{P, T} P$$

$$\Delta G = \Delta H - T\Delta S$$

$\ominus \quad \quad \quad \ominus \quad \quad \quad \oplus$

There are two criterion for spontaneity

- i) Tendency to have min. energy ($\Delta H < 0$)
- ii) " " " " max. entropy ($\Delta S > 0$)

<u>Sr. No.</u>	<u>ΔH</u>	<u>ΔS</u>	<u>ΔG</u>	<u>Comment</u>
1.	$\ominus$	$\oplus$	$\ominus$	spontaneous
2.	$\oplus$	$\ominus$	$\oplus$	non-spontaneous
3.	$\oplus$	$\oplus$	$\ominus$	spontaneous at $\uparrow T$
4.	$\oplus$	$\oplus$	$\oplus$	non-spontaneous at $\downarrow T$
5.	$\ominus$	$\ominus$	$\oplus$	non-spontaneous at $\uparrow T$
6.	$\ominus$	$\ominus$	$\ominus$	spontaneous at $\downarrow T$

* Calculation of ΔG $\Rightarrow$

1] For Physical Process :

Δ In physical process, P & T cannot be constant simultaneously. As a result sign of ΔG will not predict anything about spontaneity.

$$G = H - TS$$

$$G = U + PV - TS$$

$$dG = dU + PdV + VdP - TdS - SdT$$

$$dG = \cancel{dU} - \cancel{PdV} + \cancel{PdV} + VdP - \cancel{TdS} - SdT$$

$$dG = VdP - SdT$$

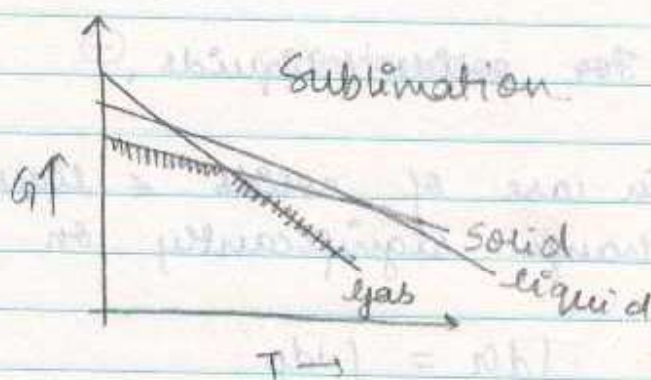
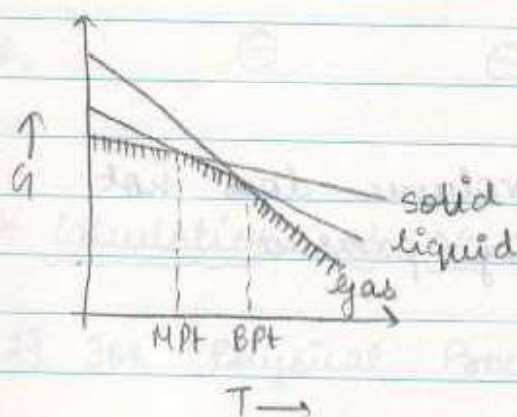
Applicable only for physical process.

→ At low pressure, G of gas is lowest whereas at high pressure all substances except water tend to be in solid state which has lowest G .

c Case II: Variation of G with T at constant pressure

$$dG = -SdT$$

$$\left(\frac{\partial G}{\partial T}\right)_P = -S$$



At low temp., solid is more stable & at $\uparrow T$, gas is more stable.

2) For Chemical Reaction $\rightarrow$



$$\begin{array}{cccc} \Delta_f G^\circ & \Delta_f G^\circ & \Delta_f G^\circ & \Delta_f G^\circ \\ \Delta_f H^\circ & \Delta_f H^\circ & \Delta_f H^\circ & \Delta_f H^\circ \end{array}$$

$$\Delta_{\text{ran}} G^\circ = d\Delta_f G^\circ(D) + c\Delta_f G^\circ(C) - a\Delta_f G^\circ(A) - b\Delta_f G^\circ(B)$$

$$\Delta_{\text{ran}} H^\circ = \sum (\nu_i \Delta_f H_i^\circ)_{\text{product}} - \sum (\nu_i \Delta_f H_i^\circ)_{\text{reactants}}$$

For any element in their reference state, Gibbs free energy of formation is taken as 0.

$$\Delta_f G^\circ (H_2(g)) = 0$$

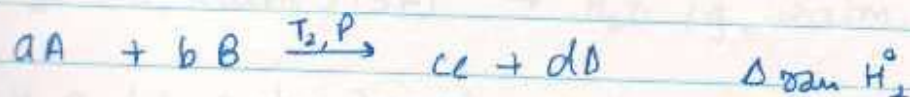
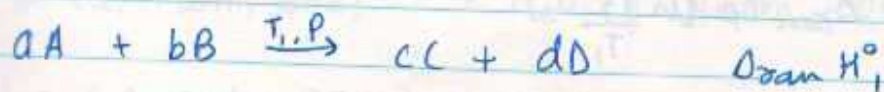
$$\Delta_f G^\circ (O_2(g)) = 0$$

$$\Delta_f G^\circ (CO_2(g)) \neq 0$$

$$\Delta_f G^\circ (O_3(g)) \neq 0$$

$$\Delta_f G^\circ (C(\text{graphite})) = 0$$

3) Variation of $\Delta_{\text{ran}} H^\circ$ with temperature :



$$dH = n C_{p,m} dT$$

for 1 mol of Σ_{an} ,

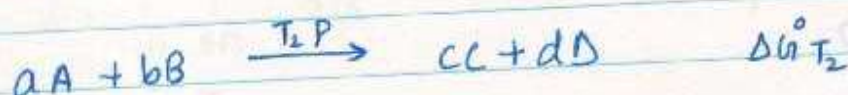
$$\int_{\Delta H_{T_1}}^{\Delta H_{T_2}} d(\Delta H) = \int_{T_1}^{T_2} \Delta \Sigma_{\text{an}} C_p dT$$

$$\boxed{\Delta H_{T_2} = \Delta H_{T_1} + \Delta \Sigma_{\text{an}} C_p (T_2 - T_1)} \rightarrow \text{Kirchoff's eq}^n$$

Similarly

$$\Delta U_{T_2} = \Delta U_{T_1} + \Delta \Sigma_{\text{an}} C_p (T_2 - T_1)$$

b) Variation of ΔG° with temperature :



$$\boxed{\Delta G_{T_2}^\circ = \Delta H_{T_2}^\circ - T_2 \Delta S_{T_2}^\circ}$$

where,

$$\Delta H_{T_2} = \Delta H_{T_1} + \Delta \Sigma_{\text{an}} C_p (T_2 - T_1)$$

$$\Delta S_{T_2} = \Delta S_{T_1} + \Delta \Sigma_{\text{an}} C_p \ln \frac{T_2}{T_1}$$

19/10

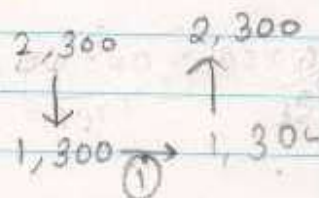
g. Find sign of ΔG

- | | <u>ΔG</u> |
|---|------------------------------|
| i) $H_2O(l, 1 \text{ atm}, 373 \text{ K}) \rightarrow H_2O(g, 1 \text{ atm}, 373 \text{ K})$ | 0 |
| ii) $H_2O(l, 1 \text{ atm}, 400 \text{ K}) \rightarrow H_2O(g, 1 \text{ atm}, 400 \text{ K})$ | -ve |
| iii) $H_2O(l, 1 \text{ atm}, 300 \text{ K}) \rightarrow H_2O(g, 1 \text{ atm}, 300 \text{ K})$ | +ve |
| iv) $H_2O(l, 0.5 \text{ atm}, 373 \text{ K}) \rightarrow H_2O(g, 0.5 \text{ atm}, 373 \text{ K})$ | -ve |
| v) $H_2O(l, 2 \text{ atm}, 373 \text{ K}) \rightarrow H_2O(g, 2 \text{ atm}, 373 \text{ K})$ | +ve |
| vi) $H_2O(s, 1 \text{ atm}, 273 \text{ K}) \rightarrow H_2O(l, 1 \text{ atm}, 273 \text{ K})$ | 0 |
| vii) $H_2O(s, 1 \text{ atm}, 300 \text{ K}) \rightarrow H_2O(l, 1 \text{ atm}, 300 \text{ K})$ | -ve |
| viii) $H_2O(s, 1 \text{ atm}, 200 \text{ K}) \rightarrow H_2O(l, 1 \text{ atm}, 200 \text{ K})$ | +ve |
| ix) $H_2O(s, 0.5 \text{ atm}, 273 \text{ K}) \rightarrow H_2O(l, 0.5 \text{ atm}, 273 \text{ K})$ | +ve |
| x) $H_2O(s, 2 \text{ atm}, 273 \text{ K}) \rightarrow H_2O(l, 2 \text{ atm}, 273 \text{ K})$ | +ve |

g. Calculate ΔG for 1 mol of H_2O

- i) $H_2O(l, 1 \text{ atm}, 300 \text{ K}) \rightarrow H_2O(g, 1 \text{ atm}, 300 \text{ K})$
- ii) $H_2O(l, 2 \text{ atm}, 373 \text{ K}) \rightarrow H_2O(g, 2 \text{ atm}, 373 \text{ K})$
- iii) $H_2O(l, 2 \text{ atm}, 300 \text{ K}) \rightarrow H_2O(g, 2 \text{ atm}, 300 \text{ K})$

Given: $\Delta H_{373 \text{ K}} = 40 \text{ kJ}$
 $\Delta G(l) = 75$ $\Delta G(g) = 35$



9)

$$\rightarrow \Delta G_{300K} = \Delta H_{300K} - T \Delta S_{300K}$$

$$\Delta H_{373K} = \Delta H_{300K} + \Delta_{\text{ran}} C_p (373 - 300)$$

$$\Delta S_{373K} = \frac{\Delta H}{T=373} = \frac{40000}{373}$$

$$\Delta S_{373K} = \Delta S_{300K} + \Delta_{\text{ran}} C_p \ln \frac{373}{300}$$

* Relationship b/w G , $\Delta_{\text{ran}} G^\circ$ & K_{eq} :

$$dG = V dp - S dT$$

At constant temp.,

$$dG = V dp$$

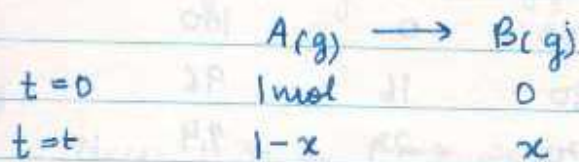
For 1 mol of an ideal gas

$$\int_{G^\circ}^{G} dG = 1 \times RT \int_{P=1\text{bar}}^{P=P_{\text{ran}}} \frac{dp}{p}$$

$$\Rightarrow G - G^\circ = 1 \times RT \ln P$$

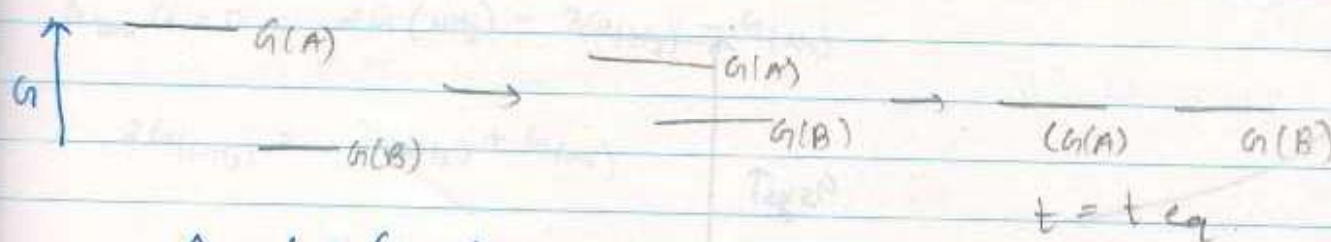
$$\therefore G(A) = n (G^\circ(A) + RT \ln P) \rightarrow \text{for } n \text{ mol}$$

Suppose we take A at 1 bar (std. state)



$$G_A = (G_A^\circ + RT \ln P_A) n_A$$

$$G_B = (G_B^\circ + RT \ln P_B) n_B$$



$$\Delta_{rxn} G = G_B - G_A$$

At equilibrium,

$$\Delta_{rxn} G = 0 \Rightarrow G_A = G_B$$

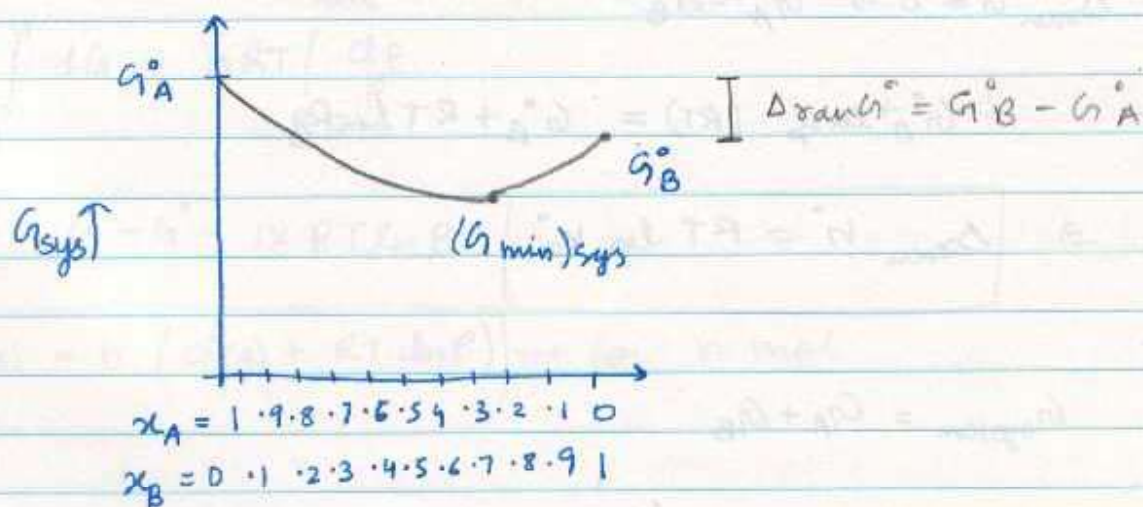
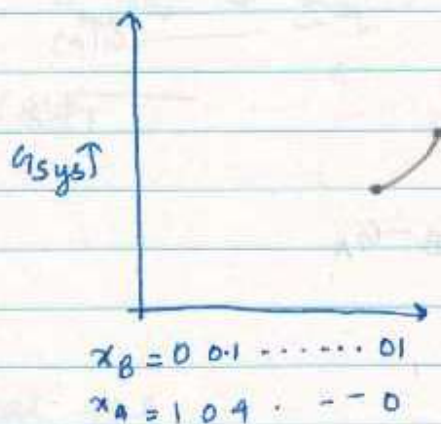
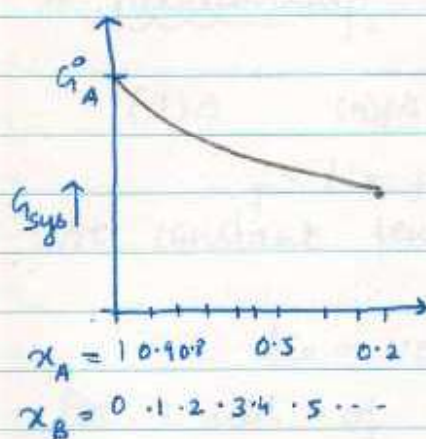
$$G_A^\circ + RT \ln P_A = G_B^\circ + RT \ln P_B$$

$$\Rightarrow \Delta_{rxn} G^\circ = RT \ln K^\circ$$

$$G_{system} = G_A + G_B$$

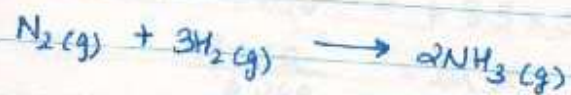
eg, $A(g) \longrightarrow B(g)$

	G_A	G_B	G_{system}
$t=0$	100	0	100
$t=t_1$	80	16	96
t_2	70	24	94
t_3			
t_4			
t_5			
$t=t_{\text{eq}}$	25	60	85



A process will move in a direction where it can $\downarrow$ or minimize Gibbs free energy of a system (its).

eg,

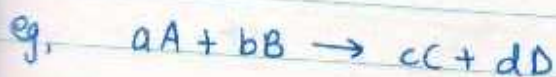


$$\Delta G_{\text{rxn}} = 2G(\text{NH}_3) - 3G(\text{H}_2) - G(\text{N}_2)$$

At eqⁿ,

$$\Delta_{\text{rxn}} G = 0 = 2G(\text{NH}_3) - 3G(\text{H}_2) - G(\text{N}_2)$$

$$2G(\text{NH}_3) = 3G(\text{H}_2) + G(\text{N}_2)$$



$$\Delta_{\text{rxn}} G = dG(D) + cG(C) - aG(A) - bG(B)$$

$$= d(G^\circ(D) + RT \ln P_D) + c(G^\circ(C) + RT \ln P_C) - a(G^\circ(A) + RT \ln P_A) - b(G^\circ(B) + RT \ln P_B)$$

$$\Delta_{\text{rxn}} G = \Delta_{\text{rxn}} G^\circ + RT \ln Q$$

At equilibrium,

$$\Delta_{\text{rxn}} G = 0 \quad \& \quad Q^\circ = K^\circ$$

$$\Delta_{\text{ran}} G^\circ = -RT \ln K$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta H^\circ - T\Delta S^\circ = -RT \ln K$$

$$\boxed{\ln K = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{R} \times \frac{1}{T}}$$

Variation of K_{eq} with temp.

We are assuming ΔH° & ΔS° as temp. independent

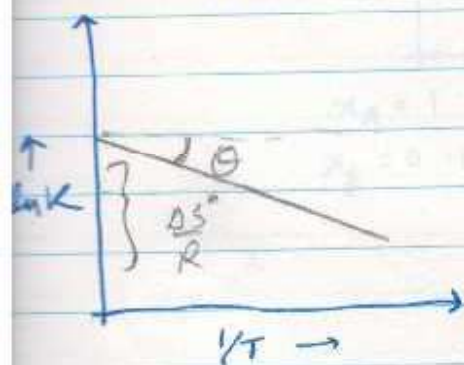
$$\ln K_{T_1}^\circ = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{R} \times \frac{1}{T_1} \quad \text{--- (I)}$$

$$\ln K_{T_2}^\circ = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{R} \times \frac{1}{T_2} \quad \text{--- (II)}$$

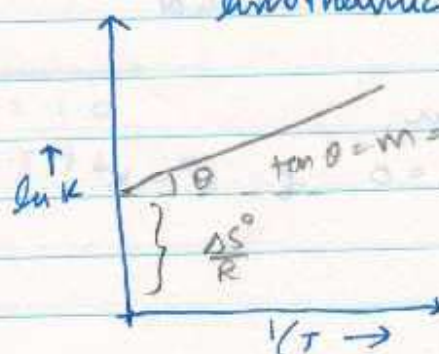
① - ①

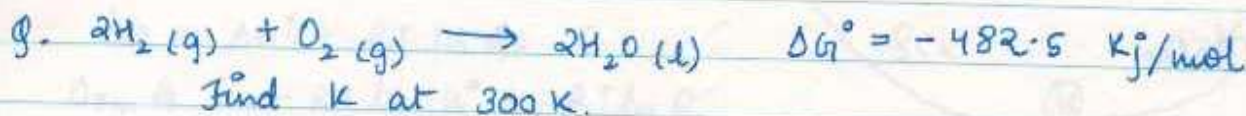
$$\ln \frac{K_{T_2}}{K_{T_1}} = \frac{\Delta H^\circ}{R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Endothermic



Endothermic





$$\rightarrow +482.5 \times 10^3 = +8.3 \times 300 \ln K$$

$$\frac{48250}{8.244} = \ln K \Rightarrow K = e^{194}$$

For rxn,



$K_{eq} = 10^{-36}$

$\Delta G^\circ (\text{kJ/mol})$

K_{eq}°

+200

8.7×10^{-36}

+100

3×10^{-18}

+50

1.7×10^{-9}

+10

6.7×10^{-1}

+1

0.95

0

1

-1

1.5

-10

5.7×10^1

-50

5.8×10^8

-100

3.4×10^{17}

-200

1.1×10^{35}

non-spontaneous

Spontaneous

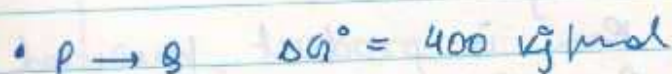
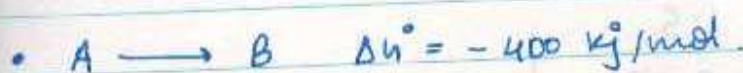
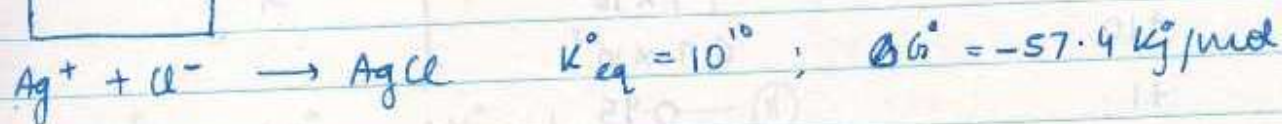
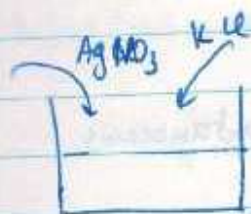
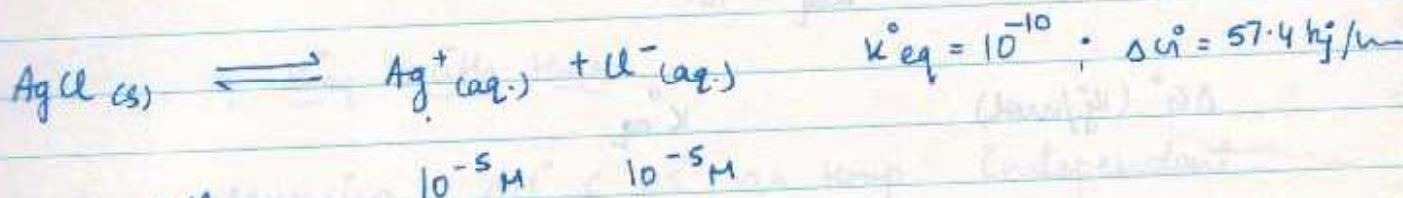
$\Delta_{rxn} G^\circ$ tells us about extent of rxn

- if $\Delta_{rxn} G^\circ < 0 \Rightarrow K_{eq}^\circ > 1 \Rightarrow$ Rxn is product favoured
- if $\Delta_{rxn} G^\circ > 0 \Rightarrow K_{eq}^\circ < 1 \Rightarrow$ Rxn is Reactant favoured
- if $\Delta_{rxn} G^\circ = 0 \Rightarrow K_{eq} = 1$

True / False

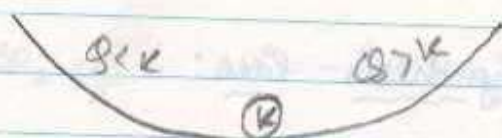
If $\Delta_{\text{ran}} G^\circ < 0 \Rightarrow$ Process must take place. (False)
cos we have to provide conditions

ΔG° tells us about potential of rxn if suitable conditions are given



$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta_{ran} G = -RT \ln K^\circ + RT \ln Q$$



$$\Delta_{ran} G = RT \ln \frac{Q^\circ}{K^\circ}$$

if $\Delta_{ran} G < 0 \Rightarrow Q^\circ < K^\circ \Rightarrow$ Ran will move in forward direction

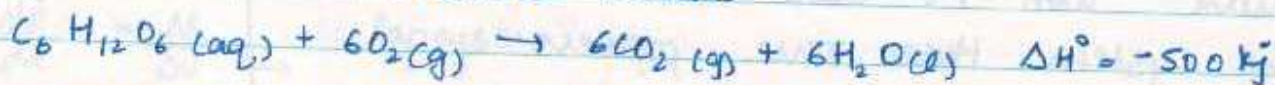
if $\Delta_{ran} G > 0 \Rightarrow Q^\circ > K^\circ \Rightarrow$ Ran will move in backward direction

if $\Delta_{ran} G = 0 \Rightarrow Q^\circ = K^\circ \Rightarrow$ Ran is in eq.

$\Delta_{ran} G$ tells us about direction of reaction.

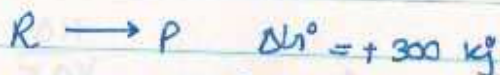
* Physical Significance of Gibbs Free Energy (G):

- It is amt of energy which is available freely for doing non-PV work or useful work like electrical work or muscle work.



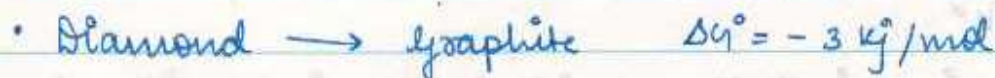
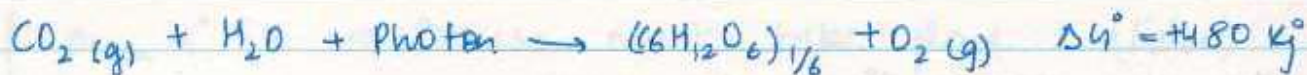
$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\rightarrow \Delta G^\circ = -300 \text{ kJ}$$



It means if 300 kJ of non PV work is done then one mole of R can be converted into 1 mol of P.

• Photosynthesis Rxn:



• $dW_{\text{non PV}} \geq dG_{P,T}$ if $\Delta G = -300 \text{ kJ}$
 $dW_{\text{by non PV}} \leq 300 \text{ kJ}$

$$dW_{\text{non PV}} = -dW_{\text{by non PV}}$$

$$-dW_{\text{by non PV}} \geq dG_{P,T}$$

for rev.,

$$dW_{\text{by non PV}} = 300 \text{ kJ}$$

$$dW_{\text{by non PV}} \leq -(dG)_{P,T}$$

for irrev.,

$$dW_{\text{non PV}} < 300 \text{ kJ}$$

Decrease in G at const. P & T will be equal to max. non PV work that can be extracted from the system thru rev. process.

* Driving non spontaneous Rxn $\rightarrow$

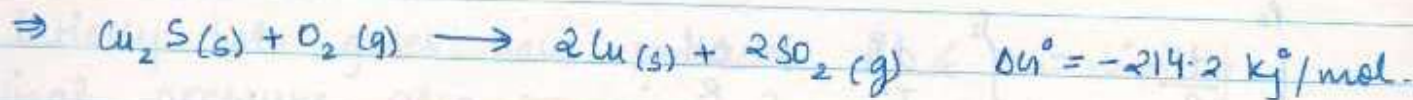
Decomposition of Cu_2S to its elements is non spontaneous rxn



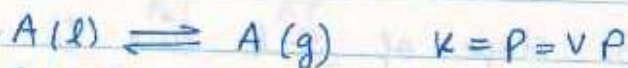
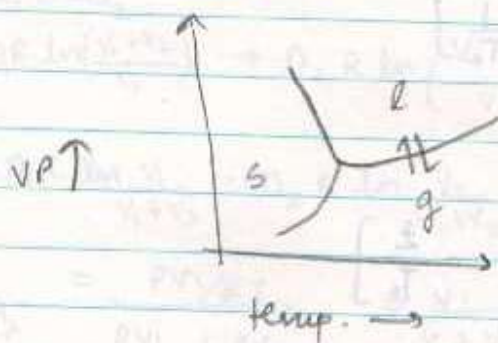
It means we cannot get pure copper directly but we can couple it with another rxn.



Ex. 1. & 2 complete



* Clapeyron equation $\Rightarrow$



at equilibrium,

$$G_A(l) = G_A(g)$$

$$dG_A(l) = dG_A(g)$$

$$V_l dp - S_l dT = V_g dp - S_g dT$$

$$(V_l - V_g) dp = (S_l - S_g) dT$$

$$\frac{dp}{dT} = \frac{S_g - S_l}{V_g - V_l} = \frac{\Delta S}{\Delta V}$$

$$\boxed{\frac{dp}{dT} = \frac{\Delta S}{\Delta V}} \rightarrow \text{Clapeyron eq.}$$

$$\Delta S = \frac{\Delta H}{T} \quad \text{during phase change}$$

$$\frac{dp}{dT} = \frac{\Delta H}{T \Delta V}$$

if we take liquid gas eq.,

$$\Delta H = \Delta_{\text{vap}} H$$

$$\frac{dp}{dT} = \frac{\Delta_{\text{vap}} H P}{RT^2}$$

$$\int_{P_1}^{P_2} \frac{dp}{P} = \int_{T_1}^{T_2} \frac{dT}{T^2} \cdot \frac{\Delta H}{R}$$

$$\Rightarrow \ln \frac{P_2}{P_1} = \frac{\Delta_{\text{vap}} H}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

For normal B.Pt.,

$$\ln \frac{P_2}{1} = \frac{\Delta_{\text{vap}} H}{R} \left[\frac{1}{T_{\text{Bpt}}} - \frac{1}{T_2} \right]$$

- In case of water, ΔG & ΔH slope of S-L line is negative.
- Slope for solid-liquid line is very steep coz ΔV for solid-liquid is very small.

* Trouton's Rule $\rightarrow$

$$A(l) \rightarrow A(g) \quad \Delta_{\text{vap}} S = \frac{\Delta_{\text{vap}} H}{T_{\text{Bpt}}} = 10.5R \approx 88 \text{ J/molK}$$

Accⁿ to it entropy of vapourisation (molar) of normal liquids at their B.Pt. is constant (10.5R or 88J)

- This rule is not applicable where H-bonding takes place like in water, ethanol, etc.
- It is just an observation.

* Entropy of mixing for ideal gases $\rightarrow$

P	P
T	T
V_1	V_2
n_1	n_2

 $\rightarrow$

P
T
$V_1 + V_2$
$n_1 + n_2$

Initially both gases have pressure P & on mixing final pressure also remains P .

$$\Delta S = \Delta S_1 + \Delta S_2$$

$$= +n_1 R \ln\left(\frac{V_1+V_2}{V_1}\right) + n_2 R \ln\left(\frac{V_1+V_2}{V_2}\right)$$

$$\Delta_{\text{mix}} S = -n_1 R \ln \frac{V_1}{V_1+V_2} - n_2 R \ln \frac{V_2}{V_1+V_2}$$

$$x_1 = \frac{n_1}{n_1+n_2} = \frac{PV_1/RT}{\frac{PV_1}{RT} + \frac{PV_2}{RT}} = \frac{V_1}{V_1+V_2}$$

$$\Delta_{\text{mix}} S = -(n_1+n_2)R [x_1 \ln x_1 + x_2 \ln x_2]$$

$$= -n_T R [x_1 \ln x_1 + x_2 \ln x_2]$$

→ we can apply it for ideal sol

Q. Find molar entropy of dry air

$$\underset{28}{N_2} = 78\%$$

$$\underset{32}{O_2} = 21\%$$

$$\underset{40}{Ar} = 1\% \quad (\text{by volume})$$

$$\rightarrow \Delta_{\text{mix}} S = -1 \times R [0.78 \ln 0.78 + 0.21 \ln 0.21 + 0.01 \ln 0.01]$$